

Bar-to-bar matchings for monomorphisms and epimorphisms

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Abstract

[Abstracts of three recent talks on this topic](#)

AATRN: Barcode matchings and p-norms of persistence modules

In one-parameter persistence, matchings can be used to relate persistence barcodes, and are involved in the definition of commonly used distances between them. A theory of matchings induced by morphisms of persistence modules, called induced matchings, has been developed by Bauer and Lesnick. In this talk, we present a different type of matchings associated with morphisms of persistence modules. After highlighting some differences with the induced matchings, we study the interaction between both types of matchings and a notion of cost for morphisms based on p-norms of persistence modules. Lastly, we discuss some consequences of our results in the context of algebraic stability of persistence barcodes.

ATMCS12: On p-norms and matchings induced by morphisms of persistence modules

Matchings between persistence barcodes play an important role in the theory of one-parameter persistence, underpinning the definition of metrics for their comparison and providing a direct way to relate them. In a foundational paper, Bauer and Lesnick developed a theory of matchings induced by morphisms of persistence modules, called induced matchings. In this talk, we introduce an alternative notion of induced matching between persistence barcodes. Our construction is based on an algebraic decomposition theorem for certain monomorphisms and epimorphisms between persistence modules. We study the interaction



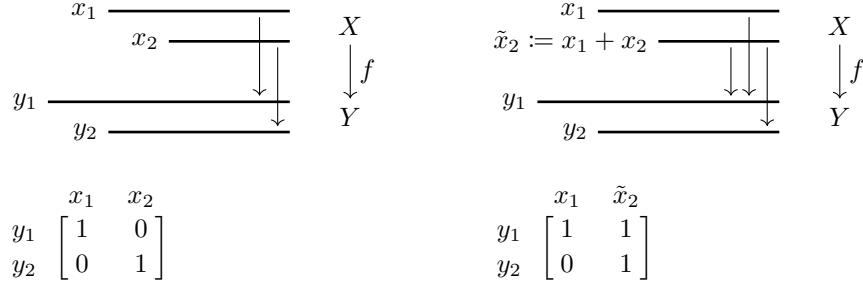
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Leibniz International Proceedings in Informatics

Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany



■ **Figure 1** Two representations of the same morphism $f: X \rightarrow Y$ in $\mathbf{Pfd}$, as bars with arrows and as matrices. On the left, the basis $\{x_1, x_2\}$ makes the monomorphism strictly bar-to-bar. This is not the case for the basis $\{x_1, \tilde{x}_2\}$ chosen on the right. **this is an example showing that the basis matter for the morphism representation**

Maybe represent the change in basis as the pre-composition by an automorphism $X \rightarrow X$.

between our induced matchings and a notion of cost for morphisms based on p-norms of persistence modules, showing that matchings can be viewed as simplified morphisms with a reduced cost. We also highlight the differences from the induced matching of Bauer and Lesnick and the similarities with a matching proposed by Gonzalez-Diaz, Soriano-Trigueros and Torras-Casas. Lastly, we illustrate some consequences of our results on the cost of morphisms and matchings in the context of algebraic stability of persistence barcodes, including a simple proof of the algebraic Wasserstein stability of persistent homology.

London TDA: Barcode matchings, Wasserstein metrics, and applications

Matchings between persistence barcodes play a key role in persistence theory, providing tractable combinatorial ways to compare topological summaries of data. A theory of matchings induced by morphisms of persistence modules, called induced matchings, was first developed by Bauer and Lesnick. In this talk, we present an alternative notion of induced matching between persistence barcodes, derived from an algebraic decomposition theorem. We study the interaction between induced matchings and a notion of cost for morphisms based on p-norms of persistence modules, showing that matchings can be viewed as simplified morphisms with a reduced cost. This result has both theoretical and practical implications. On the theoretical side, induced matchings yield a concise proof of the algebraic Wasserstein stability of persistent homology. On the practical side, our results enable efficient computation of some stable vectorisations of persistence barcodes defined via Wasserstein metrics.

1 Introduction

we need an introduction

Not sure what to do with Figure 1.

► **Remark 1.1.** Note that this notion of persistence module is more general than the usual finitely presented or “tame” modules [16]. In this context we do have a decomposition theorem [5] (see also [10]), but we do not necessarily have a finite number of bars with the same start or end-point (note that bars can be open or closed on either end).

2 Preliminaries

2.1 Persistence modules

We fix a field K and denote by vect_K the category of finite dimensional vector spaces over K . **Viewing $(\mathbb{R}, \leq)$ as a poset category**, a **pointwise finite dimensional persistence module over K** is a functor $X: \mathbb{R} \rightarrow \text{vect}_K$, consisting therefore of a collection of finite dimensional vector spaces $\{X_t\}_{t \in \mathbb{R}}$ and linear maps between them $X_{s \leq t}$ for $s \leq t$, we call transition functions. We denote the category of pointwise finite dimensional persistence modules and natural transformations between them by Pfd . In the following, pointwise finite dimensional persistence modules are referred to as persistence modules for brevity. The category Pfd has biproducts (i.e. direct sums), which are computed pointwise¹.

In this setting, interval modules are not parameterized by the real line, but rather by an extension of the real line:

► **Definition 2.1** (Decorated real line and intervals). *This definition is adapted from [3, §2.3]. We consider the **decorated real line***

$$\mathbb{E} := \mathbb{R} \times \{-, +\} \cup \{(-\infty, +), (+\infty, -)\} \subseteq [-\infty, +\infty] \times \{-, +\}$$

equipped with the following lexicographic order: for (s, σ) and (t, τ) in $\mathbb{E}$, we have $(s, \sigma) < (t, \tau)$ if $s < t$ in $[-\infty, +\infty]$ or if $s = t$, $\sigma = -$, and $\tau = +$. We denote $(t, -)$ and $(t, +)$ in $\mathbb{E}$ by t^- and t^+ , respectively, except when $t = \pm\infty$, where we omit the sign.

Ordered pairs of elements of $\mathbb{E}$ are in one-to-one correspondence with all intervals on the real line: for $s^\sigma < t^\tau$ in $\mathbb{E}$, we define the interval

$$\langle s^\sigma, t^\tau \rangle := \{u \in \mathbb{R} \mid s < u < t \text{ or } s^\sigma = u^- \text{ or } t^\tau = u^+\}.$$

In particular, for $s < t$ in $\mathbb{R}$, we have the following equality of intervals:

$$\begin{aligned} \langle s^-, t^- \rangle &= [s, t], & \langle s^-, t^+ \rangle &= [s, t], \\ \langle s^+, t^- \rangle &= (s, t), & \langle s^+, t^+ \rangle &= (s, t]. \end{aligned}$$

► **Definition 2.2** (Overlapping intervals). *Let $a, b, c, d \in \mathbb{E}$. Following [4], we say that **the interval $\langle a, b \rangle$ overlaps the interval $\langle c, d \rangle$ above** (and antisymmetrically, $\langle c, d \rangle$ **overlaps $\langle a, b \rangle$ below**) if $c \leq a < d \leq b$.*

► **Definition 2.3** (Interval modules). *Let $a < b$ be elements of $\mathbb{E}$. A **bar** (or **interval module**) with **start-point** a and **end-point** b , denoted as $K(a, b)$, is the persistence module defined as follows: for any u in $\mathbb{R}$,*

$$K(a, b)_u := \begin{cases} K & \text{if } u \in \langle a, b \rangle, \\ 0 & \text{otherwise,} \end{cases}$$

and for any $u \leq v$ in $\mathbb{R}$,

$$K(a, b)_{u \leq v} := \begin{cases} \text{id}_K & \text{if } u, v \in \langle a, b \rangle, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, the **support** of $K(a, b)$, i.e. the real values where the functor is nonzero, is the interval $\langle a, b \rangle$.

¹ This is folklore, apparently.

► **Remark 2.4.** We observe that there exists a nonzero morphism $K(a, b) \rightarrow K(c, d)$ if, and only if, $\langle a, b \rangle$ overlaps $\langle c, d \rangle$ above.

In order to better manage these interval modules, we will be considering the following subcategory of $\mathbf{Pfd}$. Recall that a **multiset** is a set S equipped with a **multiplicity** $m_S: S \rightarrow \mathbb{N}$, and that when there is no ambiguity, S and m_S may be used interchangeably.

► **Definition 2.5** (Subcategory of bars). We denote by $\mathbf{Bar}$ the full subcategory of $\mathbf{Pfd}$ whose objects are the (potentially infinite) direct sums of bars $\bigoplus_{i \in I} K(a_i, b_i)$. We call this the **category of bars**.

Given an object $X = \bigoplus_{i \in I} K(a_i, b_i)$ of $\mathbf{Bar}$, we define its **barcode** as a representation of the multiset $\mathcal{B}(X) := \{\langle a_i, b_i \rangle\}_{i \in I}$ of intervals appearing in X .

► **Definition 2.6.** Let X be a persistence module in $\mathbf{Pfd}$. A **barcode decomposition** of X is the given of an object $\bigoplus_i K(a_i, b_i)$ in $\mathbf{Bar}$ and an isomorphism $\varphi_X: X \cong \bigoplus_i K(a_i, b_i)$.

► **Remark 2.7.** The decomposition theorem [10, 5] states that every persistence module X in $\mathbf{Pfd}$ admits a barcode decomposition $\varphi_X: X \cong \bigoplus_i K(a_i, b_i)$. While the isomorphism φ_X is not unique (give an example), the direct sum of bars $\bigoplus_i K(a_i, b_i)$ is, up to reordering of the bars. In particular, we can speak of the **barcode** of an object in $\mathbf{Pfd}$ as well.

► **Remark 2.8.** Every persistence module in $\mathbf{Pfd}$ has at most countably many bars $K(s, t)$ with $s < t \in E$ in its barcode decomposition. In other words, up to an **ephemeral** summand (see [7]), which is a direct sum of bars of the form $K(s^-, s^+)$ supported on $\{s\}$, every persistence module X in $\mathbf{Pfd}$ has at most countably many bars. Moreover, for each $s \in \mathbb{R}$, X has a finite number of bars $K(s^-, s^+)$, since the dimension of X_s is finite. As a consequence, for all $b \in \mathbb{E}$, X only has countably many bars with start-point b , and likewise countably many with end-point b .

► **Proposition 2.9.** The categories $\mathbf{Bar}$ and $\mathbf{Pfd}$ are equivalent, given the axiom of choice.

Proof. We observe that the inclusion functor $\mathbf{Bar} \hookrightarrow \mathbf{Pfd}$ is fully faithful as the inclusion of a full subcategory and is essentially surjective by the decomposition theorem. Using the axiom of choice, we conclude that the inclusion functor is an equivalence of categories: see [18, Lemma 02C3]. ◀

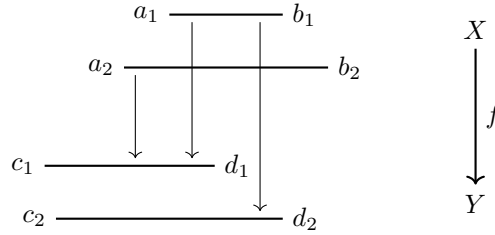
Recall from [1] that a persistence module X in $\mathbf{Pfd}$ is **tame** if there exist real numbers $0 = t_0 < t_1 < \dots < t_k$ such that the transition function $X_{s \leq t}$ is a non-isomorphism only if $s < t_i \leq t$ for some $i \in \{1, \dots, k\}$. All the bars in the barcode decomposition of a tame persistence module are of the form $\langle s^-, t^- \rangle = [s, t)$.

As a notion of *size* for persistence modules, we use the p -norm first introduced in [8] and [17], also studied in [1] for tame persistence modules.

► **Definition 2.10.** For $p \in [1, \infty]$, the p -**norm** of a persistence module with barcode decomposition $X \cong \bigoplus_i K(a_i, b_i)$ is:

$$\|X\|_p = \begin{cases} (\sum_i (b_i - a_i)^p)^{\frac{1}{p}} & \text{for } p \in [1, \infty), \\ \sup_i \{b_i - a_i\} & \text{for } p = \infty. \end{cases}$$

By Remark 2.8, the p -norm of a persistence module in $\mathbf{Pfd}$ involves an infinite sum with at most countably many positive summands. Following [17] (Definition 2.11) we say that a persistence module X has *bounded p -norm* if $\|X\|_p^p < \infty$.



■ **Figure 2** Representation of a morphism $f: K(a_1, b_1) \oplus K(a_2, b_2) \rightarrow K(c_1, d_1) \oplus K(c_2, d_2)$ in Bar . An arrow between two bars $K(a_i, b_i)$ and $K(c_j, d_j)$ corresponds to the projection of f to $\text{hom}(K(a_i, b_i), K(c_j, d_j))$; the morphism is represented by its nonzero projections.

2.2 Morphisms and matrices

One advantage of Bar over Pfd is that we can conveniently, and not just up to isomorphism, represent morphisms as (possibly infinite) matrices. Let $X = \bigoplus_{i \in I} K(a_i, b_i)$ and $Y = \bigoplus_{j \in J} K(c_j, d_j)$ be objects of Bar . Note that $\text{hom}(X, Y)$ is a vector space, possibly infinite-dimensional. Moreover, given the canonical isomorphisms

$$\text{hom}\left(\bigoplus_{i \in I} K(a_i, b_i), \bigoplus_{j \in J} K(c_j, d_j)\right) \cong \prod_{i \in I} \prod_{j \in J} \text{hom}(K(a_i, b_i), K(c_j, d_j)) \quad (1)$$

and

$$\text{hom}(K(a_i, b_i), K(c_j, d_j)) \cong \begin{cases} K & \text{if } \langle a_i, b_i \rangle \text{ overlaps } \langle c_j, d_j \rangle \text{ above,} \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

we can then represent each morphism $f \in \text{hom}(X, Y)$ with a matrix $F \in K^{I \times J}$, where $F_{ij} = 0$ if $\langle a_i, b_i \rangle$ does not overlap $\langle c_j, d_j \rangle$ above.

► **Definition 2.11** (Matrix representations of bar morphisms). *Let $f: \bigoplus_{i \in I} K(a_i, b_i) \rightarrow \bigoplus_{j \in J} K(c_j, d_j)$ be a morphism in Bar . An **ordered matrix representation** of f is $(F_{ij})_{i \in I, j \in J}$ where, for all $i \in I$ and $j \in J$, the entry F_{ij} is the projection of f to $\text{hom}(K(a_i, b_i), K(c_j, d_j))$ via the canonical isomorphisms (1) and (2), and I and J are ordered in a way compatible with the product order on the pairs (a_i, b_i) and (c_j, d_j) . **TODO: Propagate this new notion of non-unique matrix representations...***

► **Example 2.12.** For example, the **only ordered** matrix representation of the morphism represented in Figure 2 is:

$$F = \begin{matrix} & K(a_1, b_1) & K(a_2, b_2) \\ \begin{matrix} K(c_1, d_1) \\ K(c_2, d_2) \end{matrix} & \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \end{matrix}$$

Under certain conditions, the composition of morphisms can also be represented by matrices.

► **Definition 2.13** (Column-finite matrices). *A matrix $M \in K^{I \times J}$ with rows indexed by I and columns indexed by J is **column-finite** if each column only has a finite number of nonzero coefficients.*

► **Proposition 2.14.** *Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be morphisms in Bar , F and G their matrix representations. Suppose that F and G are column-finite. Then the matrix*

representation of $g \circ f$ is

$$\left(\sum_{j \in J} F_{ij} G_{j\ell} \mathbb{I}\{a_i \leq e_\ell < b_i \leq f_\ell\} \right)_{i \in I, \ell \in L},$$

where $X = \bigoplus_{i \in I} K(a_i, b_i)$, $Y = \bigoplus_{j \in J} K(c_j, d_j)$, $Z = \bigoplus_{\ell \in L} K(e_\ell, f_\ell)$, and $\mathbb{I}\{a_i \leq e_\ell < b_i \leq f_\ell\} = 1$ if the inequalities hold and 0 otherwise.

Proof. It's harder to write than to understand... ◀

- Remark 2.15. 1. The reason we assume that F and G are column-finite is to ensure that the sums are well-defined. There are other settings where the sums are well-defined, e.g. for row-finite matrices, but we will only need the result for column-finite matrices.
- 2. When every interval overlaps every other interval, composition of functions corresponds exactly to matrix multiplication. For example, this is the case when every interval has the same endpoint.

we do not discuss how elements in I and J are ordered in this subsection, should we? Also in relation to the end of remark 2.8.

2.3 Matchings

Set-theoretic **matchings** are defined as partial injective functions between sets [4], but we will use the following equivalent definition in order to more easily generalize to multiset matchings.

► **Definition 2.16** (Matchings). *Let I and J be sets. A (set-theoretic partial) matching between I and J is a subset $\mu \subseteq I \times J$ such that the projections π_I and π_J onto I and J , respectively, are injective. We denote by $\text{Match}(I, J)$ the set of matchings between I and J .*

We use the definitions of the category **Mch** (including composition of matchings $\circ$), **multiset representations**, **barcodes**, and the category **Barc** (including the overlap composition of matchings $\bullet$) from [4]. We also define multiset matchings:

► **Definition 2.17** (Multiset matchings [12, Definition 2.4]). *Let (S, m_S) and (T, m_T) be two multisets. A (partial multiset) matching between S and T is a multiset (M, m_M) where $M = S \times T$ and, for all $s \in S$ and $t \in T$,*

$$\sum_{t' \in T} m_M(s, t') \leq m_S(s) \quad \text{and} \quad \sum_{s' \in S} m_M(s', t) \leq m_T(t).$$

We denote by $\text{Match}(S, T)$ the set of matchings between S and T .

In particular, this definition coincides with the set-theoretic one when S and T are sets (m_S and m_T bounded above by 1).

► **Definition 2.18** (Matching costs). *Let S and T be multisets of intervals of the real line, $M \in \text{Match}(S, T)$ a matching, and $p \in [1, \infty]$. The p -cost of M is*

$$c_p(M) := \left(\sum_{(\langle s^\sigma, t^\tau \rangle, \langle s'^{\sigma'}, t'^{\tau'} \rangle) \in M} m_M(\langle s^\sigma, t^\tau \rangle, \langle s'^{\sigma'}, t'^{\tau'} \rangle) (|s - s'|^p + |t - t'|^p) \right)^{\frac{1}{p}}.$$

In other words, $c_p(M)$ is the p -norm of the vector of lengths of intervals in the symmetric differences of matched intervals in M .

2.4 Dualization of persistence modules and matchings

In this paper, we will prove results for monomorphisms in $\mathbf{Pfd}$. Analogous results hold for epimorphisms by a duality argument. Specifically, we consider the contravariant dualization endofunctor $(\cdot)^*: \mathbf{Pfd} \rightarrow \mathbf{Pfd}$ defined by [3, Section 3.4], which send a map $f: X \rightarrow Y$ to the **dual map** $f^*: Y^* \rightarrow X^*$. In particular, this functor sends epimorphisms to monomorphisms, and vice versa.

The dual of a matching switches the domain and codomain, switches start- and end-points, and changes the sign of start- and end-points.

3 Bar-to-bar morphisms and induced matchings

3.1 Bar-to-bar morphisms

► **Definition 3.1.** A morphism $f: X \rightarrow Y$ of persistence modules in $\mathbf{Bar}$ is **strictly bar-to-bar** if, writing $X = \bigoplus_{i \in I} K(a_i, b_i)$ and $Y = \bigoplus_{j \in J} K(c_j, d_j)$, there exists a (set-theoretic) matching $\mu \in \text{Match}(I, J)$ such that

$$f = \bigoplus_{(i,j) \in \mu} f_{ij} \oplus \bigoplus_{i \in I \setminus \pi_I(\mu)} g_i \oplus \bigoplus_{j \in J \setminus \pi_J(\mu)} h_j, \quad (3)$$

where each f_{ij} is a nonzero morphism $K(a_i, b_i) \rightarrow K(c_j, d_j)$, each g_i denotes the zero morphism $K(a_i, b_i) \rightarrow 0$ and each h_j denotes the zero morphism $0 \rightarrow K(c_j, d_j)$. The matrix associated to a strictly bar to bar morphism has at most one non zero entry for each column and one non zero entry for each row.

A morphism $f: X \rightarrow Y$ in $\mathbf{Pfd}$ is **bar-to-bar** if there exist barcode decompositions of X and Y such that the associated morphism in $\mathbf{Bar}$ is strictly bar-to-bar. In other words, there exist barcode decompositions $(\varphi_X, \bigoplus_i K(a_i, b_i))$, $(\varphi_Y, \bigoplus_j K(c_j, d_j))$ of X and Y respectively and $g: \bigoplus_i K(a_i, b_i) \rightarrow \bigoplus_j K(c_j, d_j)$ strictly bar-to-bar such that the following diagram commutes:

$$\begin{array}{ccc} \bigoplus_i K(a_i, b_i) & \xrightarrow{\varphi_X} & X \\ g \downarrow & & \downarrow f \\ \bigoplus_j K(c_j, d_j) & \xrightarrow{\varphi_Y} & Y \end{array}$$

Not every morphism is bar-to-bar: see Figure 3 for an example where no isomorphisms would make $f: X \rightarrow Y$ strictly bar-to-bar.

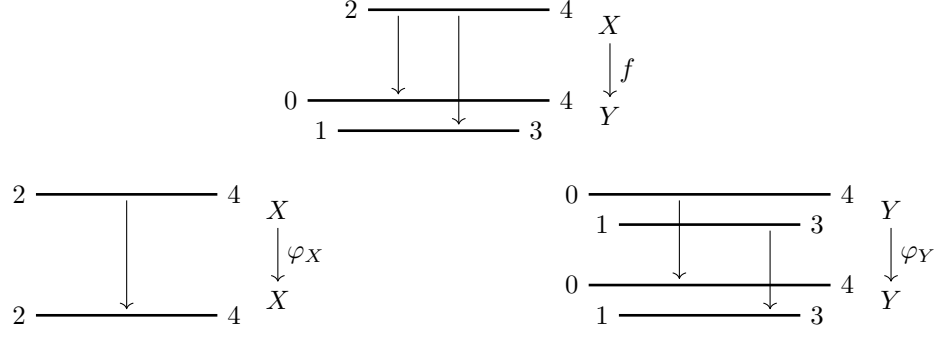
► **Proposition 3.2.** The composition of two strictly bar-to-bar monomorphisms is strictly bar-to-bar.

Proof. Consider strictly bar-to-bar morphisms $f: X \rightarrow Y$ and $f': Y \rightarrow Z$ with decompositions

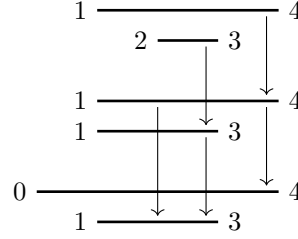
$$f = \bigoplus_{(i,j) \in \mu} f_{ij} \oplus \bigoplus_{i \in I \setminus \pi_I(\mu)} g_i \oplus \bigoplus_{j \in J \setminus \pi_J(\mu)} h_j \quad \text{and} \quad f' = \bigoplus_{(j,\ell) \in \mu'} f'_{j\ell} \oplus \bigoplus_{j \in J \setminus \pi_J(\mu')} g'_j \oplus \bigoplus_{\ell \in L \setminus \pi_L(\mu')} h'_\ell,$$

where in particular $\mu \in \text{Match}(I, J)$ and $\mu' \in \text{Match}(J, L)$ are set-theoretic matchings. Their composition $f' \circ f$ has the decomposition

$$f' \circ f = \bigoplus_{\substack{(i,\ell) \in \mu' \circ \mu \\ (i,j) \in \mu, (j,\ell) \in \mu'}} f'_{j\ell} \circ f_{ij} \oplus \bigoplus_{i \in I \setminus \pi_I(\mu)} g_i \oplus \bigoplus_{\ell \in L \setminus \pi_L(\mu')} h'_\ell,$$



■ **Figure 3** Example of a non bar-to-bar morphism $f : X \rightarrow Y$. The only valid automorphisms $\varphi_X : X \rightarrow X$, $\varphi_Y : Y \rightarrow Y$ are represented below. In both cases, the image of the restriction of the automorphism to a bar is contained within that same bar. Consequently, they do not alter the morphism when composed.



■ **Figure 4** Representation of the morphisms in Bar of Remark 3.3.

where $\mu' \circ \mu$ is the set-theoretic matching that results from composing μ and μ' as multiset matchings, and the j in the first sum is unique, given $(i, \ell) \in \mu' \circ \mu$. We observe that $f'_{j\ell} \circ f_{ij}$ is nonzero if, and only if, (i, ℓ) is in the overlap composition $\mu' \bullet \mu$ (see Remark 2.4). If not, then $i \in I \setminus \pi_I(\mu' \bullet \mu)$, $\ell \in L \setminus \pi_L(\mu' \bullet \mu)$, and $f'_{j\ell} \circ f_{ij}$ is equal to the sum of zero maps $(K(a_i, b_i) \rightarrow 0) \oplus (0 \rightarrow K(e_\ell, f_\ell))$. Thus we can write

$$f' \circ f = \bigoplus_{\substack{(i, \ell) \in \mu' \bullet \mu \\ (i, j) \in \mu, (j, \ell) \in \mu'}} f'_{j\ell} \circ f_{ij} \oplus \bigoplus_{i \in I \setminus \pi_I(\mu' \bullet \mu)} g_i \oplus \bigoplus_{\ell \in L \setminus \pi_L(\mu' \bullet \mu)} h'_\ell,$$

which is an equation of the form (3), and this concludes the proof. ◀

► **Remark 3.3.** The composition of arbitrary bar-to-bar monomorphisms is not necessarily bar-to-bar. For example, consider the following composition of morphisms (see Figure 4):

$$K(1, 4) \oplus K(2, 3) \xrightarrow{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} K(1, 4) \oplus K(1, 3) \xrightarrow{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}} K(0, 4) \oplus K(1, 3),$$

where the sign decorations of the start- and end-points are not important. Clearly the left map is bar-to-bar. The right map is bar-to-bar by doing a base change on the middle persistence module. However, the composition is a morphism between persistence modules with trivial automorphism groups, but its matrix representation looks like that of the right morphism, and therefore the composition morphism is not bar-to-bar.

We can view the matching μ of Definition 3.1 as a matching of barcodes. Specifically, μ induces the multiset matching (M, m_M) where $M = \mathcal{B}(X) \times \mathcal{B}(Y)$ and, for all $(\langle a, b \rangle, \langle c, d \rangle) \in$

M ,

$$m_M(\langle a, b \rangle, \langle c, d \rangle) := \#\{(i, j) \in \mu \mid (a_i, b_i) = (a, b) \text{ and } (c_j, d_j) = (c, d)\}.$$

It turns out that this matching is uniquely determined as soon as we have a bar-to-bar morphism in $\mathbf{Pfd}$:

- **Proposition 3.4.** *Let $f: X \rightarrow Y$ be a bar-to-bar morphism in $\mathbf{Pfd}$, and let:*
- $\varphi_X: X \cong \bigoplus_{i \in I} K(a_i, b_i)$ and $\varphi_Y: Y \cong \bigoplus_{j \in J} K(c_j, d_j)$ be barcode decompositions;
 - $g: \bigoplus_{i \in I} K(a_i, b_i) \rightarrow \bigoplus_{j \in J} K(c_j, d_j)$ be a strictly bar-to-bar morphism in $\mathbf{Bar}$ isomorphic to f (which exists by definition);
 - μ be a matching in $\mathbf{Match}(I, J)$ associated to g (see Definition 3.1).
- Then the multiset matching $M \in \mathbf{Match}_0(\mathcal{B}(X), \mathcal{B}(Y))$ induced by μ does not depend on φ_X , φ_Y , g , nor μ .

Proof. We view f as an object in the category $\mathbf{Fun}(\mathbb{R} \times \{0 < 1\}, \mathbf{vect}_K)$. As such a representation, the decomposition (3) of g becomes

$$f \cong g \cong \bigoplus_{(i,j) \in \mu} \mathbf{R}_{\langle c_j, d_j \rangle}^{\langle a_i, b_i \rangle} \oplus \bigoplus_{i \in I \setminus \pi_I(\mu)} \mathbf{I}_0^{\langle a_i, b_i \rangle} \oplus \bigoplus_{j \in J \setminus \pi_J(\mu)} \mathbf{I}_{\langle c_j, d_j \rangle}^0,$$

where the indecomposables $\mathbf{R}_*^*$ and $\mathbf{I}_*^*$ are adapted from [13, Definition 5.2] and defined as follows:

- $\mathbf{R}_{\langle c_j, d_j \rangle}^{\langle a_i, b_i \rangle}$, where $\langle a_i, b_i \rangle$ overlaps $\langle c_j, d_j \rangle$ above, is the representation of a nonzero morphism $K(a_i, b_i) \rightarrow K(c_j, d_j)$.
- $\mathbf{I}_0^{\langle a_i, b_i \rangle}$ is the representation of the zero morphism $K(a_i, b_i) \rightarrow 0$.
- Similarly, $\mathbf{I}_{\langle c_j, d_j \rangle}^0$ is the representation of the zero morphism $0 \rightarrow K(c_j, d_j)$.

By the Krull-Remak-Schmidt-Azumaya theorem [2] (see also [5]), this decomposition is unique up to permutation of the indecomposables. This implies that the matching μ is unique up to permutations of indices i, i' in I such that $a_i = a_{i'}$ and $b_i = b_{i'}$, and similarly for J . In particular, the induced matching M on barcodes is unique. The fact that M is an overlap matching is clear from its construction, and this concludes the proof. ◀

3.2 The same-endpoint operator

We define the same-endpoint operator on direct sums of bars, and then show that it is well-defined up to isomorphism on arbitrary persistence modules in $\mathbf{Pfd}$.

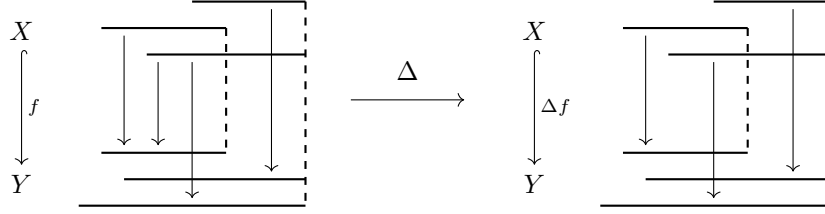
► **Definition 3.5** (same-endpoint operator Δ). *We define Δ as the operator $\mathbf{Bar} \rightarrow \mathbf{Bar}$ that is the identity on objects and the projection*

$$\mathrm{hom}(X, Y) \cong \bigoplus_{i,j} \mathrm{hom}(K(a_i, b_i), K(c_j, d_j)) \longrightarrow \bigoplus_{\substack{i,j \\ b_i = d_j}} \mathrm{hom}(K(a_i, b_i), K(c_j, d_j)) \hookrightarrow \mathrm{hom}(X, Y)$$

on morphisms, where $X = \bigoplus_i K(a_i, b_i)$ and $Y = \bigoplus_j K(c_j, d_j)$. We call Δ the **same-endpoint operator**.

► **Remark 3.6.** The first isomorphism and the last inclusion are canonical. Thus the operator Δ is well-defined and does not depend on any arbitrary choices.

► **Example 3.7.** Figure 5 shows an example of applying the same-endpoint operator to a monomorphism in $\mathbf{Bar}$.



■ **Figure 5** same-endpoint operator applied to a monomorphism $f: X \hookrightarrow Y$ in $\mathbf{Bar}$. The resulting monomorphism Δf is obtained by setting to zero all the nonzero projections between bars with different end points.

► **Proposition 3.8.** *The operator Δ is linear on hom-spaces.*

Proof. Immediate from its definition as a linear projection on hom-spaces. ◀

► **Proposition 3.9.** *The operator Δ is a functor.*

Proof. Clearly $\Delta \text{id} = \text{id}$. For compatibility with composition, consider $f: \bigoplus_i K(a_i, b_i) \rightarrow \bigoplus_j K(c_j, d_j)$ and $g: \bigoplus_j K(c_j, d_j) \rightarrow \bigoplus_k K(s_k, t_k)$ composable morphisms. We write f_{ij} for the corresponding morphism

$$\bigoplus_i K(a_i, b_i) \xrightarrow{p_i} K(a_i, b_i) \rightarrow K(c_j, d_j) \xrightarrow{t_j} \bigoplus_j K(c_j, d_j)$$

and similarly g_{jk} and $(g \circ f)_{ik}$. Then

$$g \circ f = \sum_{i,j,k} g_{jk} \circ f_{ij} = \sum_{\substack{i,j,k \\ b_i \geq d_j \geq t_k}} g_{jk} \circ f_{ij} \quad \text{and} \quad (g \circ f)_{ik} = \sum_{\substack{j \\ b_i \geq d_j \geq t_k}} g_{jk} \circ f_{ij}$$

because all other summands are 0. Moreover,

$$\Delta f = \sum_{b_i=d_j} f_{ij}, \quad \Delta g = \sum_{d_j=t_k} g_{jk}, \quad \text{and} \quad \Delta(g \circ f) = \sum_{b_i=t_k} (g \circ f)_{ik}.$$

Thus we get

$$(\Delta(g \circ f))_{ik} = \sum_{\substack{j \\ b_i \geq d_j \geq t_k \\ b_i=t_k}} g_{jk} \circ f_{ij} = \sum_{\substack{j \\ b_i=d_j=t_k}} g_{jk} \circ f_{ij} = (\Delta g \circ \Delta f)_{ik},$$

which shows the compatibility with composition. ◀

► **Corollary 3.10.** *Let $f: X \hookrightarrow Y$ be a monomorphism in $\mathbf{Bar}$. Then*

1. *If f is bar-to-bar, then it is isomorphic to Δf .*
2. *If f is strictly bar-to-bar, then $f = \Delta f$.*

Saying that two morphisms $f, g: X \rightarrow Y$ are **isomorphic** means that there exist automorphisms $\varphi: X \rightarrow X$ and $\psi: Y \rightarrow Y$ such that $f = \psi \circ g \circ \varphi$.

Proof. If f is a strictly bar-to-bar monomorphism, then it is of the form

$$\bigoplus_i (K(a_i, b_i) \hookrightarrow K(c_i, b_i)) \oplus \bigoplus_j (0 \rightarrow K(c_j, d_j))$$

where $X = \bigoplus_i K(a_i, b_i)$ and $Y = \bigoplus_i K(c_i, b_i) \oplus \bigoplus_j K(c_j, d_j)$. The second statement then follows from the definition of Δ .

For the first statement, by definition, there exist automorphisms $\varphi_X: X \rightarrow X$ and $\varphi_Y: Y \rightarrow Y$ such that $\varphi_Y \circ f \circ \varphi_X^{-1}$ is strictly bar-to-bar. Then

$$\varphi_Y \circ f \circ \varphi_X^{-1} = \Delta(\varphi_Y \circ f \circ \varphi_X^{-1}) = \Delta\varphi_Y \circ \Delta f \circ (\Delta\varphi_X)^{-1}.$$

Since Δ is a functor, it sends isomorphisms to isomorphisms. Thus Δf is isomorphic to f . $\blacktriangleleft$

► **Remark 3.11.** The converse of the first statement is also true, but it follows from Theorem 3.15. The converse of the second statement is not true: see the right side of Figure 1, where $f = \Delta f$ but f is not strictly bar-to-bar.

► **Proposition 3.12.** *Let $f: X \hookrightarrow Y$ be a monomorphism in Bar . Then Δf is also a monomorphism.*

Proof. Write $X = \bigoplus_i K(a_i, b_i)$ and $Y = \bigoplus_j K(c_j, d_j)$. Recall that Δf is a direct sum of morphisms $\bigoplus_{b_i=b} K(a_i, b_i) \rightarrow \bigoplus_{d_j=b} K(c_j, d_j)$ for all $b \in \mathbb{E}$. Thus it suffices to prove, for all $b \in \mathbb{E}$, that Δf restricted to $X^{(b)} := \bigoplus_{b_i=b} K(a_i, b_i)$ is a monomorphism.

Fix $b \in \mathbb{E}$ and write $g := \Delta f|_{X^{(b)}}$. There exists $t \in \mathbb{R}$ such that $t^- < b$ and every bar $K(a_i, b)$ in X satisfies $a_i \leq t^-$. If there were an infinite number of bars $K(c_j, d_j)$ in Y such that $t \in \langle c_j, d_j \rangle$, then Y_t would be infinite-dimensional. Since Y is in Pfd , this is not possible, so we conclude that there are only finitely many such bars, and therefore there exists $u \geq t$ in $\mathbb{R}$ such that f maps bars $K(a_i, b_i)$ in X with $b_i = b$ exclusively to bars $K(c_j, d_j)$ in Y with $d_j = b$. Thus, at u , the same-endpoint operator does not affect $f|_{X^{(b)}}$: in other words, $g_u = (f|_{X^{(b)}})_u$. In particular, $g_u = (f|_{X^{(b)}})_u$ is injective.

Next, we check that g_s is injective for all $s < u$. Indeed, if g_s sent a nonzero element $x \in X_s^{(b)}$ to 0 in Y_s , then by the transition maps we would get $X_{s \leq u}(x) \neq 0$ and $g_t(X_{s \leq u}(x)) = 0$, contradicting the injectivity of g_u . For $s \in \langle u^+, b \rangle$, injectivity holds because $g_s = (f|_{X^{(b)}})_s$ by the same argument as before. Thus g is pointwise injective, so it is a monomorphism. This concludes the proof. $\blacktriangleleft$

► **Proposition 3.13.** *Let $f: X \rightarrow Y$ be a morphism in Pfd . Let $\varphi_X: X \cong \bigoplus_i K(a_i, b_i)$ and $\varphi_Y: Y \cong \bigoplus_j K(c_j, d_j)$ be barcode decompositions of X and Y . Define $\Delta f := \varphi_Y^{-1} \Delta(\varphi_Y f \varphi_X^{-1}) \varphi_X$, where we identify the morphism $\varphi_Y f \varphi_X^{-1}$ in Pfd with the one in Bar . Then, up to isomorphism, Δf does not depend on the choice of φ_X and φ_Y .*

Proof. Let ψ_X and ψ_Y be other choices of isomorphisms. Then

$$\begin{aligned} \psi_Y^{-1} \Delta(\psi_Y f \psi_X^{-1}) \psi_X &= \psi_Y^{-1} \Delta(\psi_Y \varphi_Y^{-1} \varphi_Y f \varphi_X^{-1} \varphi_X \psi_X^{-1}) \psi_X \\ &\cong \Delta(\underbrace{\psi_Y \varphi_Y^{-1}}_{\in \text{Bar}} \underbrace{\varphi_Y f \varphi_X^{-1}}_{\in \text{Bar}} \underbrace{\varphi_X \psi_X^{-1}}_{\in \text{Bar}}) \\ &= \Delta(\psi_Y \varphi_Y^{-1}) \Delta(\varphi_Y f \varphi_X^{-1}) \Delta(\varphi_X \psi_X^{-1}) \quad (\text{by functoriality in Bar}) \\ &\cong \Delta(\varphi_Y f \varphi_X^{-1}) \\ &\cong \varphi_Y^{-1} \Delta(\varphi_Y f \varphi_X^{-1}) \varphi_X. \end{aligned}$$

► **Remark 3.14.** In particular, this means that we can define the same-endpoint operator Δ on Pfd up to isomorphism.

■ **Algorithm 1** Infinite column-finite matrix reduction. Matrix indices start from 1, I denotes the infinite identity matrix, $E_{i,j}$ denotes the infinite matrix with all 0's except a 1 in position (i,j) , and $\text{low}(j) := \max\{i \geq 1 \mid A_{i,j} \neq 0\}$.

Input: an infinite column-finite matrix M .
Output: a factorization $A = UMV$ where U and V are upper triangular invertible and A has at most one nonzero entry in each row and column.

```

 $U \leftarrow I;$ 
 $V \leftarrow I;$ 
 $A \leftarrow M;$ 
for  $j = 1, 2, \dots$  do
    while  $\exists j' < j, \text{low}(j') = \text{low}(j)$  do      // reduce by columns with same low
         $\lambda \leftarrow A_{\text{low}(j),j} / A_{\text{low}(j'),j'};$ 
         $A_{*,j} \leftarrow A_{*,j} - \lambda A_{*,j'} = A(I - \lambda E_{j',j});$ 
         $V \leftarrow V(I - \lambda E_{j',j});$ 
    end
     $i \leftarrow \text{low}(j);$ 
    while  $\exists i' < i, A_{i',j} \neq 0$  do      // reduce nonzero entries above low
         $\lambda \leftarrow A_{i',j} / A_{i,j};$ 
         $A_{i',*} \leftarrow A_{i',*} - \lambda A_{i,*} = (I - \lambda E_{i',i})A;$ 
         $U \leftarrow (I - \lambda E_{i',i})U;$ 
    end
end
return  $A, U, V$ 

```

► **Theorem 3.15.** *For all morphisms f in Pfd , the morphism Δf is bar-to-bar.*

► **Remark 3.16.** In Pfd , the morphism Δf is defined only up to isomorphism, see Proposition 3.13. Since the property of being bar-to-bar is an invariant of isomorphism classes of morphisms (by definition), Theorem 3.15 is well-formulated.

To show Theorem 3.15, we will be working with infinite matrices. In particular with infinite column-finite matrices (see Definition 2.13).

In [15] the author presents a reduction algorithm for infinite row-finite matrices (defined analogously to column-finite ones). We proceed similarly with Algorithm 1: compare with the standard algorithm of persistent homology, see [9].

Proposed change in Algorithm 1. Inside the first ‘while’, I suggest replacing the instruction $A_{*,j} \leftarrow A_{*,j} - \lambda A_{*,j'} = A(I - \lambda E_{j',j})$ by just $A_{*,j} \leftarrow A_{*,j} - \lambda A_{*,j'}$, possibly with a comment on the side saying that it is equivalent to $A \leftarrow A(I - \lambda E_{j',j})$. This is to avoid mixing column vectors and matrices. Same in the second ‘while’.

► **Lemma 3.17.** *Algorithm 1 is well-specified: it takes as input an infinite column-finite matrix M and returns a factorization $A = UMV$ where U and V are upper triangular invertible and A has at most one nonzero entry in each row and column.*

Proof. We observe that after the j^{th} step, the first j columns of V are fixed (they will not be modified in subsequent iterations). To show that the limit matrix V is invertible, we construct its left inverse iteratively, starting with $V^{-1} = I$ and then applying $V^{-1} \leftarrow (I - \lambda E_{j',j})V^{-1}$ every time we set $V \leftarrow V(I - \lambda E_{j',j})$. After the j^{th} step, the row transformations on V^{-1} involve adding rows with index $k > j$ to other rows: this will never affect the first j rows, as V^{-1} is always upper triangular.

For U , the argument is a little subtler. For all j , there exists a positive integer J such that, in the fully reduced matrix A , for all $j' > J$, $\text{low}(j') > \text{low}(j)$. Thus, after the J^{th} step, the j^{th} column of U is no longer affected by the row transformations that update U . To show that the limit matrix U is invertible, we construct its right inverse iteratively, starting with $U^{-1} = I$ and then applying $U^{-1} \leftarrow U^{-1}(I - \lambda E_{i',i})$ every time we set $U \leftarrow (I - \lambda E_{i',i})U$. After the J^{th} step, the only columns that get transformed are those with index $> \text{low}(j)$, so in particular the j^{th} column is no longer affected by the column transformations that update U^{-1} .

Finally, for A , we use the same arguments as for V and U to conclude that A converges to a well-defined limit matrix. ◀

Proof of Theorem 3.15. It suffices to prove the result for f in **Bar**, so we assume this. In this case, the morphism $\Delta f: X \rightarrow Y$ is a direct sum of morphisms between the direct summands of X and Y where all bars share a common end-point. Δf is bar-to-bar if each summand is bar-to-bar, so, without loss of generality, we can assume that X and Y are persistence modules where all bars share a common end-point. Following Remark 2.8, we can then represent Δf as an infinite matrix M with countably many columns and rows, and where the columns and rows correspond to bars and are sorted by start-point.

We observe that every column of M has a finite number of nonzero entries. Indeed, if the column corresponding to a bar $K(a, b)$ of X had an infinite number of nonzero entries, then $K(a, b)$ would map to an infinite sequence $(K(c_i, b))_i$ of bars of Y , with $c_i \leq a$ for all i . Then (since we do not consider empty bars), there exists a point $t \in \langle a, b \rangle \subseteq \langle c_i, b \rangle$ for all i , and so Y_t is infinite-dimensional, which contradicts the definition of **Bar**.

Thus we can apply Algorithm 1 to obtain invertible upper triangular matrices U and V such that UMV has at most one nonzero entry in each row and column: this corresponds to a strictly bar-to-bar morphism from X to Y . Since U and V are upper triangular, they correspond to automorphisms of X and Y , respectively. Therefore the morphism f is isomorphic to a strictly bar-to-bar morphism. In other words, f is bar-to-bar. ◀

3.3 Induced matchings

We use the definitions of **multiset representations**, **barcodes**, and the category **Barc** from [4]. While for their canonically induced matchings compose naturally, we need more structure on our persistence modules to make the matchings compose in an intuitive way.

► **Definition 3.18** (Matching operators). A **matching operator** is a function μ that takes a morphism $f: X \rightarrow Y$ in **Bar** and returns a matching (i.e., morphism) $\mu(f): \mathcal{B}(X) \rightarrow \mathcal{B}(Y)$ in **Barc**. The second argument is sometimes omitted. The operator μ is

1. **additive** if, given morphisms f and g , $\mu(f \oplus g) = \mu(f) \sqcup \mu(g)$.
2. **split** if, for a strictly bar-to-bar morphism f , $\mu(f)$ is the matching of Definition 3.1².
3. **functorial on monomorphisms** (resp. **epimorphisms**) if, given composable monomorphisms (resp. epimorphisms) f and g , $\mu(g \circ f) = \mu(g) \bullet \mu(f)$.

- **Remark 3.19.** 1. Matching operators are well-defined. In particular, they are equivariant under reindexing of the barcode multiset representations [4].
2. By [3, Proposition 5.10], such operators are never functorial over all morphisms. Moreover, the **canonically induced matching** of that paper is functorial on monomorphisms and epimorphisms.

² more precisely, the definition's matching is $\{(i, j) \mid ((\langle a_i, b_i \rangle, i), (\langle c_j, d_j \rangle, j)) \in \mu(f)\}$.

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3. Note that being split is equivalent to being additive and mapping morphisms $K(a, b) \rightarrow K(c, d)$ between single bars nontrivially.
4. **To do: we can define arbitrary additive matchings by choosing a matching for each indecomposable morphism.**

We now show that being split and functorial on monomorphisms (resp. epimorphisms) are mutually exclusive.

► **Proposition 3.20.** *A matching operator cannot be split and functorial on monomorphisms (resp. epimorphisms).*

Proof. We proceed by contradiction: suppose that such an operator μ exists. Let $b > a_1 > a_2 > \dots > a_6$ be elements of $\mathbb{E}$, and consider the sequence

$$X = K(a_1, b) \oplus K(a_2, b) \xrightarrow{f} Y = K(a_3, b) \oplus K(a_4, b) \xrightarrow{g} Z = K(a_5, b) \oplus K(a_6, b)$$

where the two monomorphisms are represented by the labeled matrices

$$f = \begin{matrix} & a_2 & a_1 \\ a_4 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ a_3 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}, \quad g = \begin{matrix} & a_4 & a_3 \\ a_6 & \begin{bmatrix} 0 & 1 \end{bmatrix} \\ a_5 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}, \quad \text{and so} \quad gf = \begin{matrix} & a_2 & a_1 \\ a_6 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ a_5 & \begin{bmatrix} 1 & 1 \end{bmatrix} \end{matrix},$$

where the label a_i corresponds to the interval $\langle a_i, b \rangle$. Recall that isomorphisms of morphisms correspond to certain left-to-right column transformations and bottom-to-top row transformations of their matrix representations. Adding the left column to the right column in the matrix representation of gf , we obtain the isomorphism

$$gf \cong \begin{matrix} & a_2 & a_1 \\ a_6 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ a_5 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix} = \underbrace{\begin{matrix} & a_4 & a_3 \\ a_6 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ a_5 & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{matrix} \circ \begin{matrix} & a_2 & a_1 \\ a_4 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ a_3 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}}_{=: h_0}. \quad (4)$$

In other words, gf and $h_0 f$ are isomorphic. Moreover, we can decompose f as

$$\begin{matrix} & a_2 & a_1 \\ a_4 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ a_3 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix} = \underbrace{\begin{matrix} & a_4 & a_3 \\ a_4 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ a_3 & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{matrix}}_{=: h_1} \circ \underbrace{\begin{matrix} & a_2 & a_1 \\ a_4 & \begin{bmatrix} 0 & 1 \end{bmatrix} \\ a_3 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}}_{=: h_2}$$

The morphism h_1 is an isomorphism, and by splitness $\mu(h_2)$ is also an isomorphism, so by functoriality (recall also that, by functoriality, μ sends isomorphisms to isomorphisms), we deduce that $\mu(f) = \mu(h_1) \circ \mu(h_2)$ is an isomorphism. Applying μ to (4), by functoriality, we get $\mu(g) \circ \mu(f) \cong \mu(h_0) \circ \mu(f)$. As we just showed that $\mu(f)$ is invertible, we deduce that $\mu(g) \cong \mu(h_0)$. But the only automorphisms of $\mathcal{B}(Y)$ and $\mathcal{B}(Z)$ are the identity matchings, so $\mu(g) = \mu(h_0)$. But this contradicts the assumption that μ is split, since g and h_0 are different strictly bar-to-bar morphisms. We conclude that there is no induced matching representation operator that is both split and functorial on monomorphisms. The epimorphism case holds by a dual argument. ◀

We can now define our induced matching. Given a morphism $f: X \rightarrow Y$ in $\mathbf{Bar}$, we know by Theorem 3.15 that Δf is bar-to-bar. Specifically, Algorithm 1 induces an overlap matching $\mu_\Delta(f) \in \mathbf{Barc}(\mathcal{B}(X), \mathcal{B}(Y))$. However, this matching is not very informative. Indeed, it is empty as soon as no bars of X or Y have the same end-point. Instead, we may follow the strategy of [3]: given a morphism $f: X \rightarrow Y$ in $\mathbf{Bar}$, consider an epi-mono factorization of f ,

$$X \xrightarrow{e} \text{im}(f) \xrightarrow{m} Y,$$

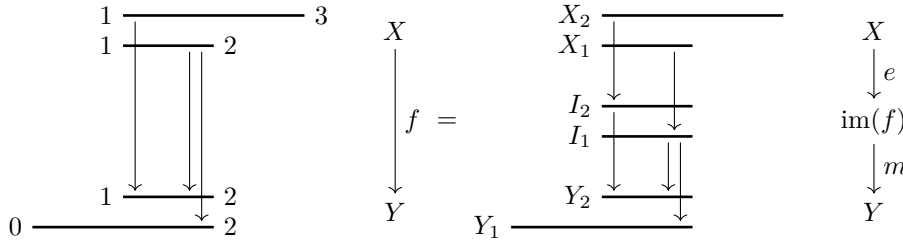
where we define $\text{im}(f)$ as the object in $\mathbf{Bar}$ that is isomorphic to the image of f in $\mathbf{Pfd}$, i.e. it is the sum of the bars in the barcode of the image. We would then define the **matching induced by f** to be the matching

$$\mu(f) := \mu_\Delta(m) \bullet \mu_\Delta(e^*)^* \quad (5)$$

in $\mathbf{Barc}$, where $-^*$ is the dual operator for epimorphisms and matchings (see Section 2.4).

The issue with this construction is that the epi-mono factorization is only unique up to isomorphism: there exist other factorizations, which are all of the form $(\varphi^{-1} \circ e, m \circ \varphi)$ with φ an automorphism on $\text{im}(f)$. Indeed, we have the following example, which shows that the induced matching depends on the choice of factorization.

► **Example 3.21.** Consider the following morphism (over the field $\mathbb{F}_2$) along with an epi-mono factorization:



where, on the right, we have named all of the bars. The image $\text{im}(f)$ consists of two bars I_1 and I_2 with identical support $\langle 1, 2 \rangle$. We have the matrix representations

$$F = \begin{matrix} & X_1 & X_2 \\ Y_1 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ Y_2 & \begin{bmatrix} 1 & 1 \end{bmatrix} \end{matrix}, \quad E = \begin{matrix} & X_1 & X_2 \\ I_1 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ I_2 & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{matrix},$$

$$\text{and } M = \begin{matrix} & I_1 & I_2 \\ Y_1 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ Y_2 & \begin{bmatrix} 1 & 1 \end{bmatrix} \end{matrix} \cong \begin{matrix} & I_1 & I_2 \\ Y_1 & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ Y_2 & \begin{bmatrix} 1 & 1 \end{bmatrix} \end{matrix} \begin{matrix} & I_1 & I_2 \\ I_1 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ I_2 & \begin{bmatrix} 0 & 1 \end{bmatrix} \end{matrix} = \begin{matrix} & I_1 & I_2 \\ Y_1 & \begin{bmatrix} 1 & 1 \end{bmatrix} \\ Y_2 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}$$

for f , e , and m , respectively (recall that we are working over the field $\mathbb{F}_2$). The induced matching is

$$\begin{aligned} \mu_\Delta(m) \bullet \mu_\Delta(e^*)^* &= \{(I_1, Y_2), (I_2, Y_1)\} \bullet \{(X_1, I_1), (X_2, I_2)\} \\ &= \{(X_1, Y_2), (X_2, Y_1)\}. \end{aligned}$$

Now consider $\varphi: \text{im}(f) \rightarrow \text{im}(f)$ the automorphism that permutes the two bars, with matrix representation

$$\Phi = \begin{matrix} & I_1 & I_2 \\ I_1 & \begin{bmatrix} 0 & 1 \end{bmatrix} \\ I_2 & \begin{bmatrix} 1 & 0 \end{bmatrix} \end{matrix}.$$

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Since $\Phi^{-1} = \Phi$, we compute

$$\Phi^{-1}E = \begin{matrix} & X_1 & X_2 \\ \begin{matrix} I_1 \\ I_2 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix} \quad \text{and} \quad M\Phi = \begin{matrix} & I_1 & I_2 \\ \begin{matrix} Y_1 \\ Y_2 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \end{matrix} \cong \begin{matrix} & I_1 & I_2 \\ \begin{matrix} Y_1 \\ Y_2 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}$$

and find the induced matching

$$\begin{aligned} \mu_{\Delta}(m \circ \varphi) \bullet \mu_{\Delta}((\varphi^{-1} \circ e)^*) &= \{(I_1, Y_2), (I_2, Y_1)\} \bullet \{(X_1, I_2), (X_2, I_1)\} \\ &= \{(X_1, Y_1), (X_2, Y_2)\} \\ &\neq \mu_{\Delta}(m) \bullet \mu_{\Delta}(e^*)^*. \end{aligned}$$

Thus the induced matching as defined in Equation (5) is not unique.

► **Lemma 3.22.** *In the category **Bar**, let $f: X \rightarrow Y$ be a morphism and $\varphi: X \rightarrow X$ and $\psi: Y \rightarrow Y$ automorphisms. Then there exist automorphic matchings $\sigma: \mathcal{B}(X) \rightarrow \mathcal{B}(X)$ and $\tau: \mathcal{B}(Y) \rightarrow \mathcal{B}(Y)$ such that*

$$\mu_{\Delta}(\psi \circ f \circ \varphi) = \tau \bullet \mu_{\Delta}(f) \bullet \sigma.$$

Proof. The morphisms f and $\psi \circ f \circ \varphi$ are isomorphic to each other. By Proposition 3.13, $\Delta(f)$ and $\Delta(\psi \circ f \circ \varphi)$ are isomorphic. By definition of μ_{Δ} and the arguments of the proof of Proposition 3.4 (this can be detailed: recall that the induced matching is read from the ladder decomposition), the induced matchings $\mu_{\Delta}(f)$ and $\mu_{\Delta}(\psi \circ f \circ \varphi)$ are equal up to permutations of bars with identical support. By the overlap condition for composition in **Barc**, automorphic matchings are exactly the bijective matchings that, at most, only permute of bars with identical support. We conclude that there exist automorphic matchings $\sigma: \mathcal{B}(X) \rightarrow \mathcal{B}(X)$ and $\tau: \mathcal{B}(Y) \rightarrow \mathcal{B}(Y)$ such that

$$\mu_{\Delta}(\psi \circ f \circ \varphi) = \tau \bullet \mu_{\Delta}(f) \bullet \sigma. \quad \blacktriangleleft$$

► **Example 3.23.** The automorphic matchings depend on the morphism f . Consider the following example. Let f and $g: X \rightarrow Y$ be morphisms in **Bar** represented by the matrices

$$F = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \cong \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad G = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

respectively, where the two bars of X have the same support, and the two bars of Y have the same support (but not necessarily the same as that of X). Let φ be the automorphism on X represented by

$$\Phi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$F\Phi = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cong \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad G\Phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Reading the pivots of the relevant matrices, we find that $\mu_{\Delta}(f) = \mu_{\Delta}(f\varphi)$ but $\mu_{\Delta}(g) \neq \mu_{\Delta}(g\varphi)$. In other words, the automorphic matchings are not the same (phrase this better).

23 Feb. To ‘keep track’ of copies of identical bars in $\text{im } f$, a ‘rule’ (e.g., longest-to-longest following matchings $\mu(e)$ and $\mu(m)$) does not always seem to return the desired matching (the Spanish one): if in the counterexample we set to zero the entry of F corresponding to (Y_2, X_1) , we would like the matching to be $\{(X_1, Y_1), (X_2, Y_2)\}$, but the longest-to-longest rule would give $\{(X_1, Y_2), (X_2, Y_1)\}$. Maybe what we want is reducing first the monomorphism m into sums of indecomposables $\oplus_b m^{(b)}$, fix a basis of $\text{im } f$ which gives this decomposition, and then ‘do something’ to reduce the matrix representing e and read out a matching. Note that we can swap the roles of m and e .

A note about the narrative of the paper: If we figure out a way of recovering the Spanish matching from the epi-mono factorization of a general morphism f , describing it would be a nice contribution, as it would provide an alternative explicit construction which is arguably less obscure than the original one. An important observation to make in the paper is that using the epi-mono factorization to define a matching carries the problem of choosing a basis for $\text{im } f$: Bauer and Lesnick don’t have that problem as their construction does not depend on bases; we circumvent the problem by choosing one basis (to do: make sure that Isaac’s algorithmic approach to defining the matching corresponds to this). Another important observation is that, to apply our main results on p -norms and p -costs of morphisms and matchings, solving the problem of ‘keeping track’ of identical bars in $\text{im } f$ is actually not necessary, as the matchings resulting from any choice have all the same kernel and cokernel (up to isomorphism).

► **Example 3.24.** We can define an induced matching on Pfd as soon as we determine a strategy for selecting, for a given morphism f in Pfd , an epi-mono factorization $f = me$. As an example, we recover the induced matchings of [11, 12] via epi-mono factorizations.

Let $f: X \rightarrow Y$ be a morphism in Bar . Denote by F the matrix representation of f where rows are ordered (lexicographically) by end-point then start-point and the columns by start-point then end-point. For each column in F , reduce it by columns to its left. This is a finite process because F is column-finite. This gives us $R = FV$ with R a matrix with unique pivots and V upper triangular and invertible.

Suppose that f is tame (needed to apply the results that give the image barcode), and write $X = \bigoplus_{j=1}^n K(a_j, b_j)$ and $Y = \bigoplus_{i=1}^m K(c_i, d_i)$. Note the inversion of indices, chosen for compatibility with matrix index notation. By [12, Theorem 5.1] (see also [19, Proposition 3.17]), the barcode of $\text{im}(f)$ consists of, for each pivot (i, j) in R whose column is labeled by $\langle a_j, b_j \rangle$ and whose row is labeled by $\langle c_j, d_i \rangle$, a bar of support $\langle a_j, d_i \rangle$.

Let I be the subset of indices in $\{1, \dots, m\}$ such that row i of R contains a pivot. Let E be a matrix whose columns are labeled like F and whose rows are indexed by I . We label row i of E by the interval $\langle a_j, d_i \rangle$, corresponding to the pivot (i, j) of R . We set

$$E_{ij} := \begin{cases} R_{ij} & \text{if } i = \text{low}(j) \text{ in } R, \\ 0 & \text{otherwise.} \end{cases}$$

We see that E represents an epimorphism from X to $\text{im}(f)$.

Let M be a matrix that solves $ME = R$ (this determines the labeling of M). Let us show that this solution exists. By construction, E has at most one nonzero coefficient in each column. Therefore, a given unknown $M_{i,i'}$ in M appears in at most one equation:

- If there exists $j \in \{1, \dots, n\}$ such that $i' = \text{low}(j)$ in R , then $E_{i',j}$ is nonzero and $M_{i,i'} = M_{i,i'}$ must satisfy the equation

$$M_{i,i'} E_{i',j} = R_{ij}.$$

If $R_{ij} = 0$, then we set $M_{i,i'} = 0$. Otherwise, we need to check that we can set $M_{i,i'}$ to be nonzero. We know that $i \leq i'$ ($R_{i',j}$ is the pivot, so R_{ij} must be above it). Column i' of M is labeled by $\langle a_j, d_{i'} \rangle$ and row i of M is labeled by $\langle c_i, d_i \rangle$. We have $d_{i'} \geq d_i$ because $i' \geq i$ and we assume that the $\langle c_i, d_i \rangle$ are ordered compatibly with the product order. We also have $a_j \geq c_i$ because R_{ij} is nonzero (and so $\langle a_j, b_j \rangle$ must overlap $\langle c_i, d_i \rangle$ above). Thus we have checked that $\langle a_j, d_{i'} \rangle$ overlaps $\langle c_i, d_i \rangle$ above, so we can set $M_{i,i'}$ to be nonzero.

■ Otherwise, $M_{i,i'}$ has no equations to satisfy, and we choose to set $M_{i,i'} = 0$.

We observe that the resulting M is in reduced row echelon form, with pivots at the coordinates (i, i) for i in I , and it represents a monomorphism (need some details here).

We thus obtain $F = MEV^{-1}$, giving us an epi-mono factorization $f = me$ by the morphisms e and m represented by EV^{-1} and M , respectively.

We define the induced matching $\mu(f) := \mu_\Delta(m) \bullet \mu_\Delta(e^*)^*$. By the last observation on M , we deduce that the induced matching $\mu_\Delta(m)$ matches each bar $\langle a_j, d_i \rangle$ in the image to $\langle c_i, d_i \rangle$ in $\mathcal{B}(Y)$. In other words, we know the matching as soon as we know $\mu_\Delta(e^*)^*$, or as soon as we know the reduced matrix R (up to permutation of identical bars). TODO: From this we deduce that this matching coincides with that of [12] (details below, in Section 5).

TODO: uniqueness of matching?

Proving uniqueness of matchings: The result [14, Lemma 2] is a generalization of the Pairing Uniqueness Lemma [9, Sect. 3]. Informally, it says that, as long as we reduce a matrix using only left-to-right column operations and bottom-up row operations, the pairing does not depend on the reduction. I think this might be useful to prove uniqueness of matchings.

We can show that the matching operator μ_Δ is additive and split (should we detail this or just observe it?). As a consequence, it is not functorial on monomorphisms nor epimorphisms.

From this operator, we can also define a unique multiset matching associated to a morphism in $\mathbf{Pfd}$:

► **Definition 3.25** (Induced multiset matching). *Let $f: X \rightarrow Y$ be a morphism in $\mathbf{Pfd}$, and $g \in \mathbf{Bar}$ isomorphic to f . We define the **multiset matching by f** to be the multiset matching $M(f)$ represented by $\mu(g)$.*

► **Lemma 3.26.** *The induced multiset matching M is well-defined, i.e. it does not depend on the choice of g .*

Proof. Let $g = me$ be the epi-mono factorization defined above somewhere.... All other morphisms in $\mathbf{Bar}$ isomorphic to f are of the form $\varphi me\psi$ with φ and ψ automorphisms (this might not be true!). Thus it suffices to observe the differences between $\mu_\Delta(m)$ and $\mu_\Delta(\varphi \circ m)$; the differences between $\mu_\Delta(e^*)$ and $\mu_\Delta((e \circ \psi)^*) = \mu_\Delta(\psi^* \circ e^*)$ are found by a dual argument. By Lemma 3.22, there exists an automorphic matching σ on X such that

$$\mu_\Delta(\varphi \circ m) = \sigma \bullet \mu_\Delta(m).$$

The only automorphic matchings are those that permute bars with identical support. We deduce that the matchings $\mu_\Delta(\varphi \circ m)$ and $\mu_\Delta(m)$ only differ by permutation of bars with identical support in the codomain $\mathcal{B}(X)$. Proceeding similarly for e , we conclude that the induced multiset matchings are identical, and therefore M is well-defined. ◀

4 Application to Wasserstein distances

4.1 Sizes and ranks of (co)kernels

In this section we prove some results in the category $\mathbf{Pfd}$. In particular, we highlight a connection between rank functions and p -norms, which yields new proofs of results in [1, Section 3].

► **Proposition 4.1.** *Let $f: X \hookrightarrow Y$ be a monomorphism in $\mathbf{Pfd}$ and consider Δf the functorially defined bar-to-bar morphism and f_φ the bar-to-bar monomorphism induced by the canonical matching between X and Y (this assumes that we fix barcodes). Then, for all $s \leq t$ in $[0, \infty)$,*

$$\mathrm{rk}((\mathrm{coker} f)_{s \leq t}) \geq \mathrm{rk}((\mathrm{coker} \Delta f)_{s \leq t}) \geq \mathrm{rk}((\mathrm{coker} f_\varphi)_{s \leq t}).$$

We first prove a series of lemmas.

► **Lemma 4.2.** *Let $f: X \hookrightarrow Y$ be an arbitrary monomorphism in $\mathbf{Pfd}$. For all $s \leq t$ in $[0, \infty)$,*

$$\mathrm{rk}((\mathrm{coker} f)_{s \leq t}) = \mathrm{rk}(Y_{s \leq t} + f_t) - \dim X_t,$$

where $Y_{s \leq t} + f_t: Y_s \oplus X_t \rightarrow Y_t$.

Proof. Consider the following diagram of vector spaces:

$$\begin{array}{ccccc} X_s & \xrightarrow{f_s} & Y_s & \longrightarrow & (\mathrm{coker} f)_s \\ X_{s \leq t} \downarrow & & Y_{s \leq t} \downarrow & & \downarrow (\mathrm{coker} f)_{s \leq t} \\ X_t & \xrightarrow{f_t} & Y_t & \longrightarrow & (\mathrm{coker} f)_t \\ & & \downarrow & & \downarrow \\ & & \mathrm{coker} Y_{s \leq t} & \longrightarrow & C \end{array}$$

where C is the cokernel of the map $(\mathrm{coker} f)_{s \leq t}$. We observe (by a diagram chase, for instance) that C is also the cokernel of the map $g := Y_{s \leq t} + f_t: Y_s \oplus X_t \rightarrow Y_t$. Thus $\dim C = \dim Y_t - \mathrm{rk} g$, and we conclude the proof by observing that

$$\begin{aligned} \mathrm{rk}((\mathrm{coker} f)_{s \leq t}) &= \dim(\mathrm{coker} f)_t - \dim C \\ &= \dim Y_t - \dim X_t - \dim Y_t + \mathrm{rk} g \\ &= \mathrm{rk} g - \dim X_t. \end{aligned}$$

◀

► **Lemma 4.3.** *Let $g: X \rightarrow Y$ be an arbitrary morphism in $\mathbf{Pfd}$. Then, for all t in $[0, \infty)$,*

$$\mathrm{rk} g_t \geq \mathrm{rk}(\Delta g)_t.$$

Proof. Fix barcode decompositions of X and Y . Then we can represent g_t as a (finite) matrix, where the columns and rows correspond to bars of X and Y that are nonzero at t , and they are sorted such that the death time is increasing. The matrix M of g_t is then block upper triangular, where the blocks correspond to bars with the same death time. The matrix M_Δ of $(\Delta g)_t$ is the block diagonal part of this matrix.

Let us show that the rank of M_Δ is smaller than the rank of M . Indeed, for every block of M_Δ , select a maximal set of linearly independent columns. The corresponding columns in M are also linearly independent, and therefore the linear map sending each of the selected columns of M_Δ to the corresponding column in M is injective. This defines an injective map $\text{im } M_\Delta \hookrightarrow \text{im } M$. Taking the dimension of both sides, we get $\text{rk}(\Delta g)_t \leq \text{rk } g_t$. ◀

► **Lemma 4.4.** *Let $f: X \hookrightarrow Y$ be an arbitrary monomorphism in Pfd . For all $s \leq t$ in $[0, \infty)$,*

$$\text{rk}(Y_{s \leq t} + f_t) \geq \text{rk}(Y_{s \leq t} + (\Delta f)_t),$$

where $Y_{s \leq t} + f_t$ and $Y_{s \leq t} + (\Delta f)_t: Y_s \oplus X_t \rightarrow Y_t$.

Proof. Since we are concerned with ranks, which are invariant by isomorphisms of morphisms, we can assume without loss of generality that X , Y , and f are in Bar .

Consider $Y[-\delta]$, the persistence module defined as $Y[-\delta]_a := Y_{a-\delta}$. Then there exists a natural morphism in Pfd $Y_{\bullet-\delta \leq \bullet}: Y[-\delta] \rightarrow Y$. The image of this morphism then has the same barcode as Y , except that every birth time has been increased by δ , with resulting bars with a birth time greater than the death time being removed. Denote by $\iota: \text{im } Y_{\bullet-\delta \leq \bullet} \hookrightarrow Y$ the monomorphism through which $Y_{\bullet-\delta \leq \bullet}$ factors. Then $\text{rk}(Y_{s \leq t} + f_t) = \text{rk}(\iota_t + f_t)$. Moreover, ι is strictly bar-to-bar, so by Corollary 3.10 $\Delta \iota = \iota$. By linearity of Δ (Proposition 3.8), we deduce that $\Delta(\iota + f) = \iota + \Delta f$. We conclude as follows:

$$\begin{aligned} \text{rk}(Y_{s \leq t} + f_t) &= \text{rk}(\iota + f)_t \\ &\geq \text{rk}(\Delta(\iota + f))_t && \text{(by Lemma 4.3)} \\ &= \text{rk}(\iota_t + (\Delta f)_t) \\ &= \text{rk}(Y_{s \leq t} + (\Delta f)_t). \end{aligned}$$

► **Lemma 4.5.** *Suppose that all of the bars of X and Y have the same end-point. Then*

$$\text{rk}((\text{coker } f_\varphi)_{s \leq t}) = (\text{rk } Y_{s \leq t} - \dim X_t)_+,$$

where $(-)_+$ is the positive part function.

Proof. Denote by b this common end-point. Since f_φ is induced by the canonical matching, we can assume without loss of generality that $X = \bigoplus_{i=1}^m K(a_i, b)$, $Y = \bigoplus_{i=1}^n K(c_i, b)$ with $m \leq n$ in $\mathbb{N} \cup \{\omega\}$, the a_i and c_i are ordered in nondecreasing order, and the canonical matching φ is the inclusion of $\{1, \dots, m\}$ in $\{1, \dots, n\}$. The fact that X and Y have at most countably many bars with end-point b is explained in Remark 2.8. Then

$$f_\varphi = \bigoplus_{i=1}^m (K(a_i, b) \rightarrow K(c_i, b)) \oplus \bigoplus_{j=m+1}^n (0 \rightarrow K(c_j, b)).$$

Therefore

$$\text{coker } f_\varphi = \bigoplus_{i=1}^m K(c_i, a_i) \oplus \bigoplus_{j=m+1}^n K(c_j, b).$$

If $t^- \geq b$, then the result holds trivially. Now consider the case $t^- < b$. Since the a_i and c_i are in nondecreasing order, there exist integers k_a and k_c such that

$$\{i \in \{1, \dots, m\} : a_i \leq t^-\} = \{1, \dots, k_a\} \quad \text{and} \quad \{i \in \{1, \dots, n\} : c_i \leq t^-\} = \{1, \dots, k_c\}.$$

Therefore

$$\{i \in \{1, \dots, n\} : c_i \leq s^-\} \setminus \{i \in \{1, \dots, m\} : a_i \leq t^-\} = \begin{cases} \{k_a + 1, \dots, k_c\} & \text{if } k_a < k_c, \\ \emptyset & \text{otherwise,} \end{cases}$$

and we conclude that

$$\begin{aligned} \text{rk}((\text{coker } f_\varphi)_{s \leq t}) &= |\{i \in \{1, \dots, m\} : c_i \leq s^-, t^- < a_i\} \sqcup \{j \in \{m+1, \dots, n\} : c_j \leq s^-\}| \\ &= |\{i \in \{1, \dots, n\} : c_i \leq s^-\} \setminus \{i \in \{1, \dots, m\} : a_i \leq t^-\}| \\ &= (k_c - k_a)_+ \\ &= (\text{rk } Y_{s \leq t} - \dim X_t)_+, \end{aligned}$$

where the second equality comes from the equality between the sets, and the fourth equality comes from the fact that $k_a = \dim X_t$ and $k_c = \text{rk } Y_{s \leq t}$. ◀

Proof of Proposition 4.1. We show the first inequality as follows:

$$\begin{aligned} \text{rk}((\text{coker } f)_{s \leq t}) &= \text{rk}(Y_{s \leq t} + f_t) - \dim X_t && \text{(by Lemma 4.2)} \\ &\geq \text{rk}(Y_{s \leq t} + (\Delta f)_t) - \dim X_t && \text{(by Lemma 4.4)} \\ &= \text{rk}((\text{coker } \Delta f)_{s \leq t}). && \text{(by Lemma 4.2)} \end{aligned}$$

We show the second inequality as follows. Since Δf is bar-to-bar by Theorem 3.15, its cokernel is the direct sum of the cokernels of the maps restricted to bars with a common end-point. Thus, without loss of generality, we can assume that the bars of X and Y all have the same end-point. In this case, we get

$$\begin{aligned} \text{rk}((\text{coker } \Delta f)_{s \leq t}) &= \text{rk}(Y_{s \leq t} + (\Delta f)_t) - \dim X_t && \text{(by Lemma 4.2)} \\ &\geq \max\{\text{rk } Y_{s \leq t}, \text{rk}(\Delta f)_t\} - \dim X_t && \text{(true for general sums of maps)} \\ &= \max\{\text{rk } Y_{s \leq t}, \dim X_t\} - \dim X_t && \text{(since } \Delta f \text{ is a monomorphism)} \\ &= (\text{rk } Y_{s \leq t} - \dim X_t)_+ \\ &= \text{rk}((\text{coker } f_\varphi)_{s \leq t}). && \text{(by Lemma 4.5)} \end{aligned}$$

◀

To connect rank functions to p -norms of cokernels, we define the following measurable space.

► **Definition 4.6.** Fix $p \in (1, \infty)$. We define the space $\Omega := \{(s, t) : s \leq t \in \mathbb{R}\}$ and the density function

$$g_p(s, t) := p(p-1)|t-s|^{p-2}.$$

► **Proposition 4.7.** For X a persistence module and $p \in (1, \infty)$,

$$\int_{\Omega} \text{rk } X_{s \leq t} g_p(s, t) \text{d}s \text{d}t = \|X\|_p^p.$$

Proof. We first consider the case where X is a single bar $K(a, b)$. Suppose that a and b are finite (i.e., not $-\infty$ nor $+\infty$, see 2.1). Then, for all $p \in (1, \infty)$, we compute

$$\int_{\Omega} \text{rk } K(a, b)_{s \leq t} g_p(s, t) \text{d}s \text{d}t = \int_a^b \int_s^b p(p-1)(t-s)^{p-2} \text{d}s \text{d}t$$

$$\begin{aligned}
 &= \int_a^b p(b-s)^{p-1} ds \\
 &= (b-a)^p \\
 &= \|K(a, b)\|_p^p.
 \end{aligned}$$

If a or b is not finite, then we still have the equality, as $\infty = \infty$.

The rank function $\text{rk } X$ is the sum of the the rank functions of the bars in the bar decomposition of X . In order to generalize the previous result, it suffices to check that

$$\int_{\Omega} \sum_{i=1}^{\infty} \text{rk } K(a_i, b_i)_{s \leq t} g_p(s, t) ds dt = \sum_{i=1}^{\infty} \int_{\Omega} \text{rk } K(a_i, b_i)_{s \leq t} g_p(s, t) ds dt,$$

where the sums are over non-ephemeral bars, as ephemeral bars do not contribute to the integral.

Adapting an older note by Andrea, we need some sort of convergence theorem:

If $\int_{\Omega} \text{rk}(X) d\mu < \infty$ the function $\text{rk}(X)$ is said to be μ -integrable or to be in $L^1(\mu)$. When is $\text{rk}(X)$ a μ_p -integrable function (where μ_p is the measure ‘defined above’, for some p)? If X has finitely many bars the answer seems easy: $\text{rk}(X)$ is integrable iff X has no infinite bars. It may be interesting to study this question for a larger class of persistence modules, for example pointwise finite dimensional ones. I think we can use Lebesgue’s dominated convergence theorem. For example, can we show that $\text{rk}(X)$ is μ_p -integrable iff X has bounded L^p energy (see definition in Skraba-Turner)? Recall the following useful application of Lebesgue’s dominated convergence theorem: if $(g_n)_{n \in \mathbb{N}}$ are functions with $g_n : \Omega \rightarrow [0, \infty]$ measurable for all n , then $\sum_{n=0}^{\infty} g_n : \Omega \rightarrow [0, \infty]$ is measurable, and $\int_{\Omega} \sum_{n=0}^{\infty} g_n d\mu = \sum_{n=0}^{\infty} \int_{\Omega} g_n d\mu$. ◀

► **Remark 4.8.** Let X be a persistence module. When $p = 1$, a real-valued function whose integral over $\mathbb{R}$ is equal to $\|X\|_1$ is the Hilbert function $(\dim X)(t) := \dim X_t$. I think this was observed in [6]. This case is similar to the statement of Proposition 4.7 since we can write $\dim X_t = \text{rk } X_{t \leq t}$. [There may even be a density function so that the case $p = 1$ is formally similar to Proposition 4.7, meaning that we can integrate over Ω instead of $\mathbb{R}$.] We can also point out that, as a consequence, the inequalities of Theorem 4.9 are equalities for $p = 1$, since $\dim(\text{coker } f)_t = \dim(\text{coker } \Delta f)_t = \dim(\text{coker } f_{\varphi})_t$ because each of them (same for any monomorphism) equals $\dim Y_t - \dim X_t$.

► **Theorem 4.9.** Let $f : X \hookrightarrow Y$ be a monomorphism in Pfd and f_{φ} the bar-to-bar monomorphism induced by the canonical matching between X and Y (this assumes that we fix barcodes). Then, for all $p \in [1, \infty]$,

$$\|\text{coker } f\|_p \geq \|\text{coker } \Delta f\|_p \geq \|\text{coker } f_{\varphi}\|_p.$$

Proof. For $p \in (1, \infty)$, we combine Propositions 4.1 and 4.7, along with the fact that the function $h \mapsto \int_{\Omega} h g_p$ is increasing.

For $p = 1$ and $p = \infty$, we conclude by continuity of p -norms with respect to p . ◀

Translate these inequalities using costs of matchings, ie $\|\text{coker } f\|_p \geq c_p(M) \geq c_p(\varphi)$ where M is the induced matching and φ is the canonical matching. Also, state the dual result and then the combined result: the cost of arbitrary f is greater than the cost of the canonical matching-induced f_{φ} .

4.2 Recovering isometry result for Wasserstein distances

To have a definition of the algebraic Wasserstein distance in $\mathbf{Pfd}$ we should show that $\|\cdot\|_p$ (defined using the equality in Proposition 4.7) is a noise system on $\mathbf{Pfd}$. We know it is true in $\mathbf{Tame}$.

To prove the ‘easy’ axiom of noise systems we can proceed as follows, showing a stronger statement.

► **Proposition 4.10.** *For every short exact sequence $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ it holds that $\mathrm{rk}(Y) \geq \mathrm{rk}(X) + \mathrm{rk}(Z)$.*

Proof. Let $s \leq t$ in $\mathbb{R}$ and consider the appropriate diagram with two exact rows as in the snake lemma. We have

$$\mathrm{rk} X_{s \leq t} - \mathrm{rk} Y_{s \leq t} + \mathrm{rk} X_{s \leq t} = -\dim \ker X_{s \leq t} + \dim \ker Y_{s \leq t} - \dim \ker Z_{s \leq t} \leq 0,$$

where the equality follows from $\mathrm{rk} X_{s \leq t} = \dim X_t - \dim \ker X_{s \leq t}$ and similar identities for Y and Z , together with the identity $\dim X_t - \dim Y_t + \dim Z_t = 0$ by exactness of $0 \rightarrow X_t \rightarrow Y_t \rightarrow Z_t \rightarrow 0$; the inequality above follows from the exact sequence $0 \rightarrow \ker X_{s \leq t} \rightarrow \ker Y_{s \leq t} \xrightarrow{g} \ker Z_{s \leq t} \rightarrow \mathrm{coker} g \rightarrow 0$ (easy part of the snake lemma) again by Euler characteristic. ◀

By Proposition 4.7 and monotonicity of the integral, we have the following consequence of Proposition 4.10.

► **Corollary 4.11** ([17], Remark 7.32). *If $p \in [1, \infty)$ and $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ is a short exact sequence, then*

$$(\|X\|_p^p + \|Z\|_p^p)^{\frac{1}{p}} \leq \|Y\|_p.$$

The ‘easy’ axiom of noise systems $\|X\|_p \leq \|Y\|_p$ and $\|Z\|_p \leq \|Y\|_p$ follows immediately. The case $p = \infty$ follows by taking the limit.

We have to prove the ‘difficult’ axiom of noise systems. (The novelty is proving the result for $\mathbf{Pfd}$)

► **Proposition 4.12.** *For every short exact sequence $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ it holds that $\|Y\|_p \leq \|X\|_p + \|Z\|_p$.*

Proof. I think we can use the canonical matching, a slight variation of what we do in the JACT paper. It would be very cool to find an inequality for rank functions that implies the result (even just for bar-to-bar monomorphisms), but it seems nontrivial. ◀

We provide an alternate (for one direction at least) proof to [17, Theorems 7.27-7.28].

► **Theorem 4.13.** *For any X and Y in $\mathbf{Pfd}$, we have $d_p^p(X, Y) = W_p(\mathrm{Dgm}(X), \mathrm{Dgm}(Y))$.*

Proof. $\boxed{\leq}$ Let $\varphi: \mathrm{Dgm}(X) \rightarrow \mathrm{Dgm}(Y)$ be a matching. We want to construct a span with a smaller or equal cost. Following [17, Theorem 7.28], define

$$Z := \bigoplus_{((a,b),(c,d)) \in \varphi} K(\max(a, c), \max(b, d))$$

and construct the natural span $X \leftarrow Z \rightarrow Y$. We can check that the cost of this span is exactly the cost of φ .

$\boxed{\geq}$ Let $X \xleftarrow{f} Z \xrightarrow{g} Y$ be a span. We want to construct a matching with a smaller or equal cost. First, factor the span as $X \hookrightarrow \text{im } f \hookleftarrow Z \hookrightarrow \text{im } g \hookrightarrow Y$ and replace each morphism with the morphism induced by canonical matchings. By Theorem 4.9 and its dual result, the cost of the new span $X \xleftarrow{f_b} Z \xrightarrow{g_b} Y$ is smaller than the cost of the original span.

The (strictly bar-to-bar) morphisms f_b and g_b induce matchings $\varphi_X: \text{Dgm}(X) \rightarrow \text{Dgm}(Z)$ and $\varphi_Y: \text{Dgm}(Y) \rightarrow \text{Dgm}(Z)$, respectively. Define

$$Z' := \bigoplus_{\substack{(a,b) \in \text{Dgm}(X) \\ (c,d) \in \text{Dgm}(Y) \\ \varphi_X(a,b) = \varphi_Y(c,d)}} K(\max(a, c), \max(b, d)).$$

Observe that f_b and g_b factor through Z' . This defines a new span (of strictly bar-to-bar/algebraic matching morphisms) $X \leftarrow Z' \rightarrow Y$ with a smaller cost than the previous span: essentially, for each bar $\langle e, f \rangle$ in Z , we consider its matched bars $\langle a, b \rangle$ and $\langle c, d \rangle$ in X and Y , respectively, and we move the endpoints of $\langle e, f \rangle$ as far left as possible while keeping f_b and g_b well-defined. This makes the cokernels and kernels of these two morphisms smaller.

By the discussion for the $\boxed{\leq}$ case, the cost of this span is equal to the cost of the matching $\psi = \varphi_X \circ \varphi_Y^{-1}$. Thus $\psi: \text{Dgm}(X) \rightarrow \text{Dgm}(Y)$ is a matching whose cost is smaller than the original span. This concludes the proof. $\blacktriangleleft$

One note on the definition of the combinatorial Wasserstein distance W_p : in the **Tame** case we can (and it is convenient) consider the definition we used in [1, Sect. 4.4], where the infimum is actually the minimum (for given X and Y) of a finite set of real numbers. The matchings in this definition are also ‘nice’, as they do not involve the points of the diagonal with infinite multiplicity (as in the other possible definition of W_p). For the **Pfd** case we may have to consider this more complicated definition of W_p . I am just wondering: would it be possible to define W_p using the rank functions instead? (this would be another contribution, but it is unclear what would play the role of matchings)

► **Proposition 4.14** (Interleaving distance is d_∞^∞). *If X and Y are δ -interleaved persistence modules (see [3] for the definition), then we can construct an explicit span $X \xleftarrow{f} Z \xrightarrow{g} Y$ whose ∞ -cost is at most δ , that is,*

$$\|(\|\ker f\|_\infty, \|\text{coker } f\|_\infty, \|\ker g\|_\infty, \|\text{coker } g\|_\infty)\|_\infty \leq \delta.$$

Conversely, if there is a span $X \xleftarrow{f} Z \xrightarrow{g} Y$ with ∞ -cost equal to δ , then X and Y are δ -interleaved.

Proof. To do. I still haven’t thought about this, I just think the statement makes sense and is interesting to have (if true). $\blacktriangleleft$

Some thoughts about the proof

Note: we can already say that the interleaving distance coincides with d_∞^∞ , as they both coincide with the bottleneck between persistence barcodes. Proposition 4.14 would be a more direct way to prove it.

The first statement in Proposition 4.14 seems easy to prove using Thm 6.4 or Corollary 6.6 in [3]. It means using the induced matchings, so it wouldn’t be a ‘direct’ proof, but it’s the only way I see for now.

I think Corollary 6.6 in [3] can be used to prove the converse direction too. If we have a span $X \xleftarrow{f} Z \xrightarrow{g} Y$ whose ∞ -cost is at most δ , then in particular $f: Z \rightarrow X(-\delta/2)(\delta/2) = X$ has

Algorithm 2 Matrix reduction for the matching $\mathcal{P}$

Input: a matrix F representing a morphism $f: X \rightarrow Y$, whose columns are ordered lexicographically (birth, death) and whose rows are ordered lexicographically (death, birth).

Output: a reduced matrix R with unique lowest nonzero coefficients.

$R \leftarrow F$;

for $j = 1, \dots, m$ **do**

while $\exists j' < j, \text{low}(j') = \text{low}(j)$ **do** // reduce by columns with same low

$R_{*,j} \leftarrow R_{*,j} - (R_{\text{low}(j),j} / R_{\text{low}(j'),j'}) R_{*,j'}$;

for $i = 1, \dots, n$ **do**

if i^{th} bar of Y and j^{th} bar of X are incompatible **then**

$R_{i,j} \leftarrow 0$;

end

end

end

end

return R

δ -trivial kernel and cokernel, which means by [3, Corollary 6.6] that there is an ‘interleaving inverse’ $h: X(-\delta/2) \rightarrow Z(\delta/2)$ (or, with a reparameterization, $h(\delta/2): X \rightarrow Z(\delta)$) with δ -trivial kernel and cokernel. Compose $h(\delta/2)$ with g to obtain a morphism $X \rightarrow Y(\delta)$ which we can immediately conclude it’s a δ -interleaving morphism (i.e., there exists an ‘interleaving inverse’) by composition of interleavings or using again Corollary 6.6 in [3]. Think about details like δ vs $\delta/2$, but I think it’s correct.

These are really ‘direct’ proofs as we use induced matchings, but I’m ok with that. This fact that we used (it’s exactly Corollary 6.6 in [3]) seems quite a powerful/useful result to me: a morphism $f: M \rightarrow N$ with small kernel and cokernel (2δ -trivial) has an ‘interleaving inverse’ $h: N \rightarrow M(2\delta)$ with small kernel and cokernel too. (Is there a similar statement for $p < \infty$).

5 Comparison with $\mathcal{P}$ -matchings

5.1 Preliminaries

In the next two sections, we show that our induced matching is the same as the one in [11]. In particular, this means our matching can be computed by the algorithm described in their paper [12] (see also [19]). From now on, we restrict ourselves to **Tame**, the category of tame persistence modules.

In this section, we show some preliminary results relating our matching μ with the matching $\mathcal{P}$ of [11]. First, we recall the definition of $\mathcal{P}$.

► **Definition 5.1** ([12]). *Let $f: X \rightarrow Y$ be a morphism of persistence modules. Let R be the reduced matrix produced by Algorithm 2. Then we define $\mathcal{P}(f)$ to be the matching described by the pivots of R . *Need to clarify if this is compatible with isomorphisms, or if we need to define this on Bar**

► **Lemma 5.2.** *Let $f: X \rightarrow Y$ be a monomorphism (resp. epimorphism) where all the bars of X and Y have the same endpoint (resp. startpoint). Then $\mu(f) = \mathcal{P}(f)$.*

Proof. Let us consider the case where f is a monomorphism. $\mu(f)$ is computed via Algorithm 2, except that the matrix columns are ordered lexicographically (death, birth), and there is no check for incompatible bars. Since all bars have the same endpoint, this does not change anything at all, so both algorithms have the exact same execution, and we conclude that $\mu(f) = \mathcal{P}(f)$.

We conclude similarly for epimorphisms, where now the rows are ordered lexicographically (birth, death). ◀

► **Lemma 5.3.** *Let $f: X \rightarrow Y$ be an arbitrary monomorphism or epimorphism. Then $\mathcal{P}(f) = \mathcal{P}(\Delta f)$.*

Proof. Let us consider the case where f is a monomorphism. We claim that, at every point in the execution of Algorithm 2, the lowest nonzero coefficients of the columns of R stay in the “diagonal blocks”, i.e. the blocks where the column and row labels all have the same endpoint. This is the case because if not, then the matrix R would not be representing a monomorphism. Indeed, if some block with common endpoint b does not contain all of the lowest nonzero coefficients of its columns, then there must be a column whose nonzero coefficient is higher up (recall that the rows are sorted lexicographically (death, birth)), and so a column that is 0 in the diagonal block. Then $f_{b-\varepsilon}$ is not a monomorphism, for $\varepsilon > 0$ small enough.

As a consequence, column operations executed by the algorithm always stay within the same block, and therefore its behavior will not change depending on if the input matrix represents f or Δf . We conclude that $\mathcal{P}(f) = \mathcal{P}(\Delta f)$.

We conclude similarly for epimorphisms, where we look at blocks whose column and row labels all have the same startpoint. ◀

From the above two lemmas we get the following result.

► **Proposition 5.4.** *Let $f: X \rightarrow Y$ be a monomorphism or epimorphism. Then $\mu(f) = \mathcal{P}(f)$.*

Proof. By definition of μ for monomorphisms and epimorphisms (reference needed), $\mu(f) = \mu(\Delta f)$. By Lemma 5.2 and the additivity of the matchings μ and $\mathcal{P}$, we have $\mu(\Delta f) = \mathcal{P}(\Delta f)$. By Lemma 5.3, we have $\mathcal{P}(\Delta f) = \mathcal{P}(f)$. Putting these together, we get

$$\mu(f) = \mu(\Delta f) = \mathcal{P}(\Delta f) = \mathcal{P}(f).$$

◀

Memo: we should add somewhere the definition of ‘pair of incompatible bars’, used in Algorithm 2.

Also: Introduce notations for tame persistence modules, e.g. $K(a, b)$, used in next section. definition of tame persistence modules is now added in preliminaries.

5.2 Algorithm for computing induced matchings

We now show that μ and $\mathcal{P}$ coincide for arbitrary morphisms. To show this, we first introduce some preliminary notions about a persistent form of linear algebra, adapted from [19].

► **Definition 5.5** (Persistence vector). *Given a persistence module X and $a \in [-\infty, \infty)$, we call a **persistence vector** of X a morphism $\alpha: K(a, \infty) \rightarrow X$. We denote by $\text{PVect}(X)$ the set of all persistence vectors of X .*

For a persistence vector α , we will write $\alpha \sim [a, b)$ to say that α is a morphism from $K(a, \infty)$ to X and that $b = \inf\{t \geq a \mid \alpha(t) = 0\}$. We will sometime refer a as $\text{birth}(\alpha)$ and b as $\text{death}(\alpha)$. We will also denote by Z_a the persistence vector $Z_a: K(a, \infty) \rightarrow X$ such that $Z_a(t) = 0$ for all $t \geq a$; in this case we have $Z_a \sim [a, a)$.

In [19], Torras-Casas defined how to do operations such as sum or scalar multiplication on these persistence vectors. It is mentioned that the sum can not simply be defined as $(\alpha_i + \alpha_j)(r) := \alpha_i(r) + \alpha_j(r)$ since it does not always define a natural transformation, as shown by the following example:

► **Example 5.6.** Consider $\alpha_1 \sim [0, 2)$ and $\alpha_1 \sim [1, 3)$ two persistence vectors of X . Since they are morphisms, they make the following diagrams commute:

$$\begin{array}{ccccccc}
 K(0, \infty) & & K & \xrightarrow{Id} & K & \xrightarrow{Id} & K \longrightarrow \dots \\
 \alpha_1 \downarrow & & \alpha_1(0) \downarrow & & \alpha_1(1) \downarrow & & \alpha_1(2)=0 \downarrow \\
 X & & X_0 & \xrightarrow{\varphi_{0,1}} & X_1 & \xrightarrow{\varphi_{1,2}} & X_2 \longrightarrow \dots \\
 \\
 K(1, \infty) & & 0 & \xrightarrow{0} & K & \xrightarrow{Id} & K \longrightarrow \dots \\
 \alpha_2 \downarrow & & \alpha_2(0)=0 \downarrow & & \alpha_2(1) \downarrow & & \alpha_2(2) \downarrow \\
 X & & X_0 & \xrightarrow{\varphi_{0,1}} & X_1 & \xrightarrow{\varphi_{1,2}} & X_2 \longrightarrow \dots
 \end{array}$$

For $\alpha_1 + \alpha_2$ to be a morphism from a bar to X , we need $\varphi_{0,1}(\alpha_1 + \alpha_2)(0) = (\alpha_1 + \alpha_2)(1)$. If the sum is defined pointwise, then we get $(\alpha_1 + \alpha_2)(0) = \alpha_1(0) + \alpha_2(0) = \alpha_1(0)$, since $\alpha_2(0) = 0$. Therefore, $\varphi_{0,1}(\alpha_1 + \alpha_2)(0) = \varphi_{0,1}\alpha_1(0) = \alpha_1(1)$. However, this is not equal to $(\alpha_1 + \alpha_2)(1) = \alpha_1(1) + \alpha_2(1)$, since $\alpha_2(1) \neq 0$. We do not get the desired commutativity, and therefore the sum can not be defined pointwise.

To overcome this issue, the sum of persistence vectors is defined in the following way which consists to truncate the part where it is not well-defined.

► **Definition 5.7** (Sum of persistence vectors). *Given two persistence vectors $\alpha_1: K(a_1, \infty) \rightarrow X$ and $\alpha_2: K(a_2, \infty) \rightarrow X$, the sum of α_1 and α_2 is the persistence vector $\alpha_1 \boxplus \alpha_2: K(\max(a_1, a_2), \infty) \rightarrow X$ defined by $(\alpha_1 \boxplus \alpha_2)(r) = \alpha_1(r) + \alpha_2(r)$ for any $r \in [\max(a_1, a_2), \infty)$. This defines an associative, commutative operation on persistence vectors.*

► **Definition 5.8** (Scalar semiring). *In order to extend the analogy with vector spaces, we define the **scalar semiring induced by K** as the commutative semiring $(K \cup \{\emptyset\}, +, \cdot)$ equal to the field $(K, +, \cdot)$ when restricted to K , and with the binary operations extended as follows: for all $\lambda \in K \cup \{\emptyset\}$,*

$$\lambda + \emptyset = \lambda \quad \text{and} \quad \lambda \cdot \emptyset = \emptyset.$$

In this semiring, $\emptyset$ is the additive identity, and $1_K \in K$ is the multiplicative identity. We denote the semiring by $K^\emptyset$.

► **Definition 5.9** (Scalar multiplication). *Given a persistence vector $\alpha \sim [a, b)$ and $\lambda \in K$, we define the scalar multiplication of α by λ as the persistence vector $\lambda\alpha$ defined by $(\lambda\alpha)(r) = \lambda \cdot \alpha(r)$ for $r \in [a, \infty)$. Note that if $\lambda = 0$, then $\lambda\alpha = Z_a$.*

Let $Z_{-\infty}: K(-\infty, \infty) \rightarrow X$ be the zero map, viewed as a persistence vector. Then we define $\emptyset\alpha := Z_{-\infty}$ for all α . We observe that we have defined a semimodule structure on PVect over the semiring $K^\emptyset$, with $Z_{-\infty}$ as the additive identity.

► **Definition 5.10** (Generating family). A family $\{\alpha_i \sim [a_i, b_i]\}_{i \in I}$ of persistence vectors of X is called a **generating family** of X if for any persistence vector $\gamma \sim [a, b]$ of X , there exists $(\lambda_i)_{i \in I}$ in $K^\emptyset$ such that $\gamma = (\bigoplus_{i \in I} \lambda_i \alpha_i) \boxplus Z_a$. Note that this implies $a \geq \max_{i \in I, \lambda_i \neq \emptyset} (a_i)$.

► **Definition 5.11** (Linearly independent family). We say that a family $\{\alpha_i \sim [a_i, b_i]\}_{i \in I}$ of persistence vectors is **linearly independent** if the set $\{\alpha_i(t) \mid \alpha_i(t) \neq 0\}$ is linearly independent for every $t \in \mathbb{R}$.

► **Definition 5.12** (Persistence basis). A **persistence basis** of a persistence module M is an isomorphism $\alpha: \bigoplus_i K(a_i, b_i) \rightarrow M$. The restriction of α to a single bar module $\alpha_i: K(a_i, b_i) \rightarrow M$ is called a **persistence generator**. A persistence basis is often represented by the collection of its persistence generators $\mathcal{A} = \{\alpha_i\}_i$.

► **Theorem 5.13** ([19, Proposition 3.11]). A family $\mathcal{A}$ of persistence vectors is a persistence basis if and only if $\mathcal{A}$ is a generating family and is linearly independent.

We make the following observation.

► **Proposition 5.14.** Let $f: X \rightarrow Y$ be a morphism of persistence modules. Then f induces a morphism $\text{PVect}(X) \rightarrow \text{PVect}(Y)$ that commutes with persistence sums and scalar products. That is, for $\alpha, \beta \in \text{PVect}(X)$ and $\lambda \in K \setminus \{0\}$, we have

$$f(\alpha \boxplus \lambda \beta) = f(\alpha) \boxplus \lambda f(\beta)$$

Proof. The equality is between two functions $[c, \infty) \rightarrow Y$, where c is some real number. We conclude by pointwise linearity and naturality of f . ◀

This framework gives us a new way of representing morphisms between persistence modules. **From here, I will be using the semiring approach.**

► **Definition 5.15.** Let $f: X \rightarrow Y$ be a morphism of persistence modules, $\mathcal{A} = \{\alpha_1, \dots, \alpha_m\}$ a generating family of X , and $\mathcal{B} = \{\beta_1, \dots, \beta_n\}$ a generating family of Y . The **persistence matrix representation** of f w.r.t. $\mathcal{A}$ and $\mathcal{B}$ is the following labeled matrix with coefficients in $K^\emptyset$:

$$\begin{matrix} & \alpha_1 & \cdots & \alpha_m \\ \beta_1 & \lambda_{11} & \cdots & \lambda_{1m} \\ \vdots & \vdots & & \vdots \\ \beta_n & \lambda_{n1} & \cdots & \lambda_{nm} \end{matrix} \Bigg],$$

where, for each $j \in \{1, \dots, m\}$, writing $\alpha_j \sim [a_j, b_j]$, we have $f(\alpha_j) = \bigoplus_{i=1}^n \lambda_{ij} \beta_i \boxplus Z_{a_j}$. Note that the λ_{ij} are uniquely determined only if $\mathcal{B}$ is linearly independent. (assuming we put $\emptyset$'s when no map is possible)

► **Remark 5.16.** This representation of morphisms is quite similar to that of Definition 2.11. In particular, if the generating families are persistence bases, the barcode decompositions (Definition 2.6) we choose correspond to the persistence bases, and we replace $\emptyset$ with 0, then the matrix itself would be the exact same. The difference lies in how we interpret column and row operations. Here, performing a column operation $\alpha \leftarrow \alpha_i \boxplus \lambda \alpha_j$ is allowed as soon as the resulting family $\mathcal{A} \setminus \{\alpha_i\} \cup \{\alpha_i \boxplus \lambda \alpha_j\}$ is still generating. When allowed, this operation changes the label of the α_i column to $\alpha_i \boxplus \lambda \alpha_j$, but the matrix still represents the same morphism f . Compare this with column operations on the matrices of Definition 2.11, which correspond to (isomorphic) transformations of the morphism f itself. The same remark holds for row operations.

■ **Algorithm 3** Persistence matrix reduction

Input: a persistence matrix representation F whose columns are ordered lexicographically (birth, death) and whose rows are ordered lexicographically (death, birth).

Output: a reduced matrix R with unique lowest nonzero coefficients.

$R \leftarrow F$;

for $j = 1, \dots, m$ **do**

while $\exists j' < j, \text{low}(j') = \text{low}(j)$ **do** // reduce by columns with same low

$R_{*,j} \leftarrow R_{*,j} - (R_{\text{low}(j),j} / R_{\text{low}(j'),j'}) R_{*,j'}$;

for $i = 1, \dots, n$ **do**

if i^{th} row and j^{th} column of R are incompatible **then**

$R_{i,j} \leftarrow \emptyset$;

end

end

end

end

return R

In [19], the author shows that we can obtain the persistence basis of the image of a morphism from a matrix representation of the morphism by applying Algorithm 3. In this algorithm, we define $\text{low}(j)$ as $\max\{i \in \{1, \dots, n\} \mid R_{i,j} \notin \{0, \emptyset\}\}$, so division by $R_{\text{low}(j),j}$ is always well-defined. We also say that the i^{th} row and j^{th} column of a matrix representation R are **incompatible** if the vectors associated to their labels are incompatible, i.e. the column vector cannot map nontrivially to the row vector.

► **Proposition 5.17** ([19, Proposition 3.17]). *Let $f: X \rightarrow Y$ be a morphism of persistence modules, $\mathcal{A}$ and $\mathcal{B}$ persistence bases of X and Y respectively, and F the matrix representation of f w.r.t. $\mathcal{A}$ and $\mathcal{B}$. Suppose that $\mathcal{A}$ is ordered lexicographically by birth than death and $\mathcal{B}$ is ordered lexicographically by death than birth. Let R be the reduced matrix obtained from Algorithm 3 representing f w.r.t. the generating families $\tilde{\mathcal{A}} := \{\tilde{\alpha}_1, \dots, \tilde{\alpha}_m\}$ and $\mathcal{B}$, where $\tilde{\alpha}_j = \alpha_j \boxplus \boxplus_{k=1}^{j-1} \kappa_{jk} \alpha_k$.*

Then the set of nonzero persistence vectors $\{f(\tilde{\alpha}_j)\}_j \setminus \{Z_a\}_{a \in [-\infty, \infty)}$ is a persistence basis of $\text{im}(f)$.

We write $\gamma_i := f(\tilde{\alpha}_j)$ where the pivot of $\tilde{\alpha}_j$ in R is on the i^{th} row. Denote by I the set of indices of rows of R that contain a pivot. Then $\mathcal{C} := \{\gamma_i\}_{i \in I}$ is the above persistence basis of $\text{im}(f)$. We observe that if $\alpha_j \sim [a, a']$ and $\beta_i \sim [b, b']$, then $\gamma_i \sim [a, b']$.

It should be clear that Algorithms 2 and 3 give the same matching $\mathcal{P}$, as their differences are purely technical. The rest of this section is dedicated to showing that $\mu = \mathcal{P}$. First, we show that Algorithm 3 allows us to construct an epi-mono factorization of f .

► **Proposition 5.18.** *Let $f: X \rightarrow Y$ be a morphism of persistence modules and F the matrix representation of f w.r.t. persistence bases $\mathcal{A}$ and $\mathcal{B}$. Suppose that $\mathcal{A}$ and $\mathcal{B}$ are **properly ordered**.*

Then we have the equality of matrices

$$F = M\tilde{E}U,$$

where

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- M is the matrix representation of a monomorphism $\text{im}(f) \rightarrow Y$ w.r.t. $\mathcal{C}$ and $\mathcal{B}$, and it is in upper echelon form:

$$M = \begin{matrix} & \mathcal{C} \\ \mathcal{B} & \begin{bmatrix} 1 & * & * \\ & \ddots & * \\ & & 1 \end{bmatrix} \end{matrix}$$

Specifically, its pivots are at the coordinates (β_i, γ_i) for i in I .

- $\tilde{E}$ is the matrix representation of an epimorphism $X \rightarrow \text{im}(f)$ w.r.t. $\tilde{\mathcal{A}}$ and $\mathcal{C}$ such that $R = M\tilde{E}$.
- U is the upper triangular matrix representation of the identity morphism on X w.r.t. $\mathcal{A}$ and $\tilde{\mathcal{A}}$. In particular, $F = RU$.

Proof. U is well-defined because $\mathcal{A}$ is ordered by birth, so all the summands added onto α_j in $\tilde{\alpha}_j$ can be removed. We check that this gives an upper triangular matrix.

We define $\tilde{E}$ to be the matrix w.r.t. $\tilde{\mathcal{A}}$ and $\mathcal{C}$ where

$$\tilde{E}_{ij} = \begin{cases} R_{ij} & \text{if } i = \text{low}(j) \text{ in } R, \\ \emptyset & \text{otherwise.} \end{cases}$$

Then we see that $\tilde{E}$ represents an epimorphism from X to $\text{im}(f)$: for each persistence vector γ_i , the associated persistence vector $\tilde{\alpha}_j$ surjects onto it.

Now let us solve the following matrix equation for M :

$$M\tilde{E} = R.$$

The columns of M and the rows of $\tilde{E}$ are labeled by $\mathcal{C}$: to help keep notations clear, we will be indexing these columns and rows by I , that is, not by consecutive integers but by the indices corresponding to the γ_i .

By construction, $\tilde{E}$ has at most one nonzero coefficient in each column. Therefore, a given unknown $M_{i,i'}$ in M appears in at most one equation:

- If $i = i'$, then there exists $j \in \{1, \dots, m\}$ such that $i = \text{low}(j)$ for R . By construction, $\tilde{E}_{ij} = R_{ij}$ is nonzero, and so $M_{i,i'}$ satisfies the equation $M_{i,i'}\tilde{E}_{ij} = R_{ij}$, i.e., $M_{i,i'} = 1$.
- Otherwise, if there exists $j \in \{1, \dots, m\}$ such that $i' = \text{low}(j)$ in R , then $\tilde{E}_{\text{low}(j),j}$ is nonzero and $M_{i,i'} = M_{i,\text{low}(j)}$ must satisfy the equation

$$M_{i,\text{low}(j)}\tilde{E}_{\text{low}(j),j} = R_{ij}.$$

We need to check that this coefficient of M is proper. If R_{ij} is $\emptyset$, then we are done. If not, then we know that $i < i'$ ($R_{i',j}$ is the pivot, so R_{ij} must be above it). We have

$$\text{death}(\gamma_{i'}) = \text{death}(\beta_{i'}) \geq \text{death}(\beta_i),$$

where the equality comes from the construction of $\gamma_{i'}$ and the inequality comes from the fact that $\mathcal{B}$ is sorted by death. We also have

$$\text{birth}(\gamma_{i'}) = \text{birth}(\alpha_j) \geq \text{birth}(\beta_i),$$

where the equality comes from the construction of $\gamma_{i'}$ and the inequality comes from the fact that α_j maps to β_i in R . Thus we have checked that $\gamma_{i'}$ can map into β_i , so we can set $M_{i,i'}$ to be nonzero.

■ Otherwise, $M_{i,i'}$ has no relations, and we choose to set $M_{i,i'} = \emptyset$.

We observe that the resulting M is in reduced row echelon form, with pivots at the coordinates (β_i, γ_i) for i in I , and it represents a monomorphism (need some details here).

Thus we have constructed $\tilde{E}$ and M such that $M\tilde{E} = R$ and $F = RU$. Therefore $F = M\tilde{E}U$, and this concludes the proof. ◀

We are now ready to show the equality between μ and $\mathcal{P}$. We first need to adapt the notion of matchings to persistence vectors, and maybe change the name of the matching M .

► **Theorem 5.19.** *Let $f: X \rightarrow Y$ be a morphism of persistence modules. Then $\mu(f) = \mathcal{P}(f)$.*

Proof. Let $F = M\tilde{E}U$ be the mono-epi factorization of the persistence matrix representation of f , as constructed in Proposition 5.18. Let $m: \text{im}(f) \hookrightarrow Y$ be the monomorphism represented by M and $e: X \twoheadrightarrow \text{im}(f)$ the epimorphism represented by $\tilde{E}U$. We observe that M already has unique pivots, so $\mathcal{P}(m)$ is the matching that matches each γ_i in $\mathcal{C}$ to the corresponding β_i in $\mathcal{B}$. To read the matching induced by e , it suffices to look at the matrix $\tilde{E}$, which is the reduced form of $\tilde{E}U$. Considering $\tilde{E}$ and R , we see that $(\alpha_j, \gamma_i) \in \mathcal{P}(e)$ if, and only if, $(\alpha_j, \beta_i) \in \mathcal{P}(f)$. In other words, $\mathcal{P}(f) = \mathcal{P}(m) \circ \mathcal{P}(e)$.

On the other side, by definition of μ (??), $\mu(f) = \mu(m) \circ \mu(e)$. By Proposition 5.4, we conclude that

$$\mu(f) = \mu(m) \circ \mu(e) = \mathcal{P}(m) \circ \mathcal{P}(e) = \mathcal{P}(f).$$

◀

► **Remark 5.20.** In particular, this means that Algorithm 2 computes μ .

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